

SECTION 3—CLASSIFICATION

3.0 General Principles

3.0.1 Classifying Dangerous Goods

3.0.1.1 Dangerous goods are defined as those goods which meet the criteria of one or more of nine UN hazard classes and where applicable, to one of three UN Packing Groups according to the provisions of this section. The nine classes relate to the *type of hazard* whereas the packing groups relate to the applicable *degree of danger* within the class.

3.0.1.2 Wastes must be transported under the requirements of the appropriate class considering their hazards and the criteria of the Regulations. Wastes not otherwise subject to these Regulations, but covered under the *Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989)*, may be transported under Class 9.

3.0.1.3 Many of the substances listed in Classes 1 to 9 are deemed, without additional labelling, as being environmentally hazardous.

3.0.1.4 Classification must be made by the appropriate national authority when so required or may otherwise be made by the shipper.

3.0.1.5 A shipper who has identified, on the basis of test data, that a substance listed by name in Column B of the List of Dangerous Goods, Table 4.2 meets classification criteria for a hazard class or division that is not identified in the list, may, with the approval of the appropriate national authority, consign the substance:

- (a) under the most appropriate generic or not otherwise specified (n.o.s.) entry reflecting all hazards; or
- (b) under the same UN number and name but with additional hazard communication information as appropriate to reflect the additional subsidiary hazard(s) (documentation, label, placard) provided that the primary hazard class remains unchanged and that any other transport conditions (e.g. limited quantity and packaging provisions) that would normally apply to substances possessing such a combination of hazards are the same as those applicable to the substance listed.

3.0.1.6 When consigned under an approval granted by the appropriate national authority, a copy of the approval must accompany the consignment and the approval number must be noted on the Shipper's Declaration.

Note:

When an appropriate national authority grants such approvals, it should inform the United Nations Subcommittee of Experts on the Transport of Dangerous Goods accordingly and submit a relevant proposal of amendment to the Dangerous Goods List. Should the proposed amendment be rejected, the appropriate national authority should withdraw its approval.

3.0.2 Hazard Classes

Some hazard classes are further subdivided into hazard divisions due to the wide scope of the class. The nine hazard classes and their divisions are listed below. The order in which they are numbered is for convenience and does not imply a relative degree of danger.

3.0.2.1 Class 1—Explosives

- Division 1.1—Articles and substances having a mass explosion hazard.
- Division 1.2—Articles and substances having a projection hazard but not a mass explosion hazard.
- Division 1.3—Articles and substances having a fire hazard, a minor blast hazard and/or a minor projection hazard but not a mass explosion hazard.
- Division 1.4—Articles and substances presenting no significant hazard.
- Division 1.5—Very insensitive substances having a mass explosion hazard.
- Division 1.6—Extremely insensitive articles which do not have a mass explosion hazard.

3.0.2.2 Class 2—Gases

- Division 2.1—Flammable gas.
- Division 2.2—Non-flammable, non-toxic gas.
- Division 2.3—Toxic gas.

3.0.2.3 Class 3—Flammable Liquids

This class has no sub-divisions.

3.0.2.4 Class 4—Flammable Solids; Substances Liable to Spontaneous Combustion; Substances which, in Contact with Water, Emit Flammable Gases

- Division 4.1—Flammable solids, self-reactive substances, polymerizing substances and solid desensitized explosives.
- Division 4.2—Substances liable to spontaneous combustion.
- Division 4.3—Substances which, in contact with water, emit flammable gases.

3.0.2.5 Class 5—Oxidizing Substances and Organic Peroxides

- Division 5.1—Oxidizer.
- Division 5.2—Organic peroxides.

3.0.2.6 Class 6—Toxic and Infectious Substances

- Division 6.1—Toxic substances.
- Division 6.2—Infectious substances.

3.0.2.7 Class 7—Radioactive Material

This class has no sub-divisions.

3.0.2.8 Class 8—Corrosives

This class has no sub-divisions.

3.0.2.9 Class 9—Miscellaneous Dangerous Substances and Articles, Including Environmentally Hazardous Substances

This class has no sub-divisions.

3.0.3 Packing Groups

3.0.3.1 For packing purposes, substances are assigned to the relevant packing group according to the degree of danger they present:

- Packing Group I—high danger.
- Packing Group II—medium danger.
- Packing Group III—low danger.

△ **3.0.3.2** Criteria for Packing Groups I, II and III have been developed for substances in Classes 3, 4, Division 5.1, Division 6.1 and Class 8 and are given in Subsections 3.3, 3.4, 3.5, 3.6 and 3.8. Some substances in Class 9, and liquids in Division 5.1 have been assigned to packing groups by experience rather than through the application of any technical criteria and these are shown in the List of Dangerous Goods in Subsection 4.2. Unless otherwise provided for, the UN specification packagings detailed in the packing instructions must meet the performance test requirements of the relevant packing group shown in Column E of the List of Dangerous Goods for the particular substance.

3.0.3.3 Articles are not assigned to packing groups. For packing purposes, any requirement for a specific packaging performance level is set out in the applicable packing instruction.

3.0.4 Multiple Hazards

Subsection 3.10 describes the procedure for determining which hazard takes precedence for an article or substance with more than one hazard. However, the most stringent packing group based on the different hazards must then be the packing group for the article or substance.

3.0.5 Shipper's Responsibility

The shipper is responsible for identifying and classifying all dangerous goods intended for transport by air in compliance with these Regulations. Specifically, before packing any dangerous goods for transport by air, the shipper must:

- identify, correctly and fully, all articles and substances that meet the criteria as dangerous goods within the consignment;
- classify each item of dangerous goods by determining under which of the nine classes it falls and where relevant, determining any subsidiary hazards;
- where relevant, assign each item of dangerous goods to one of the three packing groups within the assigned class or division.

3.1 Class 1—Explosives

STATE VARIATIONS: BEG-01/02/03, GBG-01, HKG-03, KWG-01, SAG-04, USG-05/16, ZAG-01

3.1.1 Definition

Class 1 comprises:

- explosive substances (a substance which is not itself an explosive but which can form an explosive atmosphere of gas, vapour or dust is not included in Class 1), except those that are too dangerous to transport or those where the predominant hazard is appropriate to another class;
- explosive articles, except devices containing explosive substances in such quantity or of such a character that their inadvertent or accidental ignition or initiation, during transport, will not cause any effect external to the device either by projection, fire, smoke, heat or loud noise (see 3.1.7); and
- articles and substances not mentioned under (a) and (b) above which are manufactured with a view to producing a practical explosive or pyrotechnic effect.

Notes:

- Class 1 is a restricted class, that is, only those explosive articles and substances that are listed in Subsection 4.2—List of Dangerous Goods may be accepted for transport. However, the appropriate authorities of the States concerned retain the right by mutual agreement to approve transport of explosive articles and substances for special purposes under special conditions. Therefore, entries have been included in Subsection 4.2—List of Dangerous Goods for "Articles, explosive, n.o.s." and "Substances, explosive, n.o.s.". It is intended that these entries be used only when no other method of operation is possible.*
- Most of the explosives listed in Subsection 4.2—List of Dangerous Goods are not normally permitted for transport by air. Information concerning them is included in this list, however transportation must be approved by the appropriate authorities of the States concerned under the provisions of 1.2.6.*
- General entries, such as "Explosive, blasting, Type A" are used to allow for the transport of new substances. In preparing these requirements, military ammunition and explosives have been taken into consideration to the extent that they are likely to be transported by civil aircraft.*

4. A number of articles and substances, which are in Class 1, are described in Appendix A of these Regulations. These descriptions are given because a term may not be well known or may be at variance with its usage for regulatory purposes. Other definitions for terms used in this Subsection are also provided in Appendix A.
5. Class 1 is unique in that the type of packaging frequently has a decisive effect on the hazard and therefore on the assignment to a particular division. The correct division is determined by use of the procedures provided in this Subsection.

3.1.2 Description

Explosive articles and substances are assigned to one of six divisions and to one of thirteen compatibility groups. Not all compatibility groups are to be found in all divisions.

Notes:

1. Only explosives in Division 1.4, compatibility group S are permitted on a passenger aircraft.
2. Only explosives in Division 1.3, compatibility groups C and G and Division 1.4, compatibility groups B, C, D, E, G and S are permitted on a cargo aircraft.

3.1.3 Divisions

Class 1 is divided into six divisions.

3.1.3.1 Division 1.1

Articles and substances having a mass explosion hazard (a mass explosion is one which affects almost the entire load virtually instantaneously).

3.1.3.2 Division 1.2

Articles and substances having a projection hazard but not a mass explosion hazard.

3.1.3.3 Division 1.3

Articles and substances having a fire hazard and either a minor blast hazard or a minor projection hazard or both, but not a mass explosion hazard. This division comprises articles and substances that:

- (a) give rise to considerable radiant heat; or
- (b) burn one after another, producing minor blast and/or projection effects.

3.1.3.4 Division 1.4

Articles and substances that present no significant hazard. This division comprises articles and substances which present only a small hazard in the event of ignition or initiation during transport. The effects are largely confined to the package and no projection of fragments of appreciable size or range is to be expected. An external fire must not cause virtually instantaneous explosion of almost the entire contents of the package.

Note:

Articles and substances in this division are placed in Compatibility Group S when they are so packaged or designed that any hazardous effects arising from accidental functioning are confined within the package unless the package has been degraded by fire, in which case all blast or projection effects are limited to the extent that they do not significantly hinder fire-fighting or other emergency response efforts in the immediate vicinity of the package.

3.1.3.5 Division 1.5

Very insensitive substances, having a mass explosion hazard, which are so insensitive that there is very little probability of initiation or of transition from burning to detonation under normal conditions of transport.

Note:

For normal conditions of transport see 5.0.4.

3.1.3.6 Division 1.6

Extremely insensitive articles which do not have a mass explosion hazard. This division comprises articles which predominantly contain extremely insensitive substances and which demonstrate a negligible probability of accidental initiation or propagation.

Note:

The hazard from articles of Division 1.6 is limited to the explosion of a single article.

3.1.4 Compatibility Groups

3.1.4.1 Goods of Class 1 are assigned to one of six divisions, depending on the type of hazard they present (see 3.1.3) and to one of thirteen compatibility groups which identify the kinds of explosive articles and substances that are deemed to be compatible. Tables 3.1.A and 3.1.B show the scheme of classification into compatibility groups, the possible hazard divisions associated with each group and the consequential classification codes.

3.1.4.2 The definitions of compatibility groups in Table 3.1.A are intended to be mutually exclusive, except for an article or substance which qualifies for Compatibility Group S. Since the criterion of Compatibility Group S is an empirical one, assignment to this group is necessarily linked to the tests for assignment to Division 1.4.

△ 3.1.4.3 Certain Division 1.4S explosives, identified by Special Provision A165 in Subsection 4.2, are subject to Test Series 6(d) of Part I of the UN *Manual of Tests and Criteria* to demonstrate that any hazardous effects arising from functioning are confined within the package. Evidence of a hazardous effect outside the package includes:

- (a) denting or perforation of the witness plate beneath the package;
- (b) a flash or flame capable of igniting such as a sheet of $80 \pm 3 \text{ g/m}^2$ paper at a distance of 25 cm from the package;

- (c) disruption of the package causing projection of the explosives contents; or
- (d) a projection which passes completely through the packaging (a projection or fragment retained or stuck in the wall of the packaging is considered as non hazardous).

The appropriate national authority may wish to take into account the expected effect of the initiator when assessing the results of the test, if these are expected to be significant when compared to the articles being tested. If there are hazardous effects outside the package, then the product is excluded from Compatibility Group S.

TABLE 3.1.A
Compatibility Group For Explosives (3.1.4)

Compati- bility Group	Hazard Division	Article or Substance to be Classified
A	1.1	Primary explosive substance
B	1.1; 1.2; 1.4	Article containing a primary explosive substance and not containing two or more effective protective features. Some articles, such as detonators for blasting, detonator assemblies for blasting and primers, cap type, are included, even though they do not contain primary explosives
C	1.1; 1.2; 1.3; 1.4	Propellant explosive substance or other deflagrating explosive substance or article containing such explosive substance
D	1.1; 1.2; 1.4; 1.5	Secondary detonating explosive substance or black powder or article containing a secondary detonating explosive substance, in each case without means of initiation and without a propelling charge or article containing a primary explosive substance and containing two or more effective protective features
E	1.1; 1.2; 1.4	Article containing a secondary detonating explosive substance, without means of initiation, with a propelling charge (other than one containing a flammable liquid or gel or hypergolic liquids)
F	1.1; 1.2; 1.3; 1.4	Article containing a secondary detonating explosive substance, with its own means of initiation, with a propelling charge (other than one containing a flammable liquid or gel or hypergolic liquids) or without a propelling charge
G	1.1; 1.2; 1.3; 1.4	Pyrotechnic substance, or article containing a pyrotechnic substance, or article containing both an explosive substance and an illuminating, incendiary, tear-or smoke-producing substance (other than a water-activated article or one containing white phosphorus, phosphide, a pyrophoric substance, a flammable liquid or gel or hypergolic liquids)
H	1.2; 1.3	Article containing both an explosive substance and white phosphorus
J	1.1; 1.2; 1.3	Article containing both an explosive substance and a flammable liquid or gel
K	1.2; 1.3	Article containing both an explosive substance and a toxic chemical agent
L	1.1; 1.2; 1.3	Explosive article or substance containing an explosive substance and presenting a special hazard (e.g. due to water activation, or the presence of hypergolic liquids, phosphides or a pyrophoric substance) and needing isolation of each type
N	1.6	Articles containing predominantly extremely insensitive substances
S	1.4	Article or substance so packed or designed that any hazardous effects arising from accidental functioning are confined within the package unless the package has been degraded by fire, in which case all blast or projection effects are limited to the extent that they do not significantly hinder or prohibit fire fighting or other emergency response efforts in the immediate vicinity of the package

Notes:

- Articles of Compatibility Groups D and E may be fitted or packed together with their own means of initiation provided that such means have at least two effective protective features designed to prevent an explosion in the event of accidental functioning of the means of initiation. Such articles and packages must be assigned to Compatibility Groups D or E.
- Articles of Compatibility Groups D and E may be packed together with their own means of initiation, which do not have two effective protective features when, in the opinion of the appropriate national authority of the State of Origin, the accidental functioning of the means of initiation does not cause the explosion of an article under normal conditions of transport. Such packages must be assigned to Compatibility Groups D or E.

TABLE 3.1.B
Scheme of Classification of Explosives, Combination of Hazard Division
with Compatibility Group (3.1.4.1)

Hazard	Compatibility Group													
	A	B	C	D	E	F	G	H	J	K	L	N	S	A-S Σ
1.1	1.1A	1.1B	1.1C	1.1D	1.1E	1.1F	1.1G		1.1J		1.1L			9
1.2		1.2B	1.2C	1.2D	1.2E	1.2F	1.2G	1.2H	1.2J	1.2K	1.2L			10
1.3			1.3C			1.3F	1.3G	1.3H	1.3J	1.3K	1.3L			7
1.4		1.4B	1.4C	1.4D	1.4E	1.4F	1.4G						1.4S	7
1.5				1.5D										1
1.6												1.6N		1
1.1–1.6 Σ	1	3	4	4	3	4	4	2	3	2	3	1	1	35

3.1.5 Classification of Explosives

3.1.5.0 For additional information regarding classification of explosives, see UN Recommendations, 2.1.3.1.4, 2.1.3.1.5 and 2.1.3.4.

3.1.5.1 Any article or substance having or suspected of having explosive characteristics must first be considered for classification in Class 1 in accordance with the procedures in 3.1.5.2 to 3.1.5.5. Goods are not classified in Class 1 when:

- (a) unless specially authorized, the transport of an explosive substance is forbidden because sensitivity of the substance is excessive;
- (b) the article or substance comes within the scope of those explosive articles and substances which are specifically excluded from Class 1 by the definition of this class; or
- (c) the article or substance has no explosive properties.

3.1.5.2 The classification of fireworks must be based on paragraph 2.1.3.5 of the UN Recommendations.

3.1.5.3 Any article or substance having or suspected of having explosive characteristics must be considered for classification in Class 1 in accordance with the tests, procedures and criteria prescribed in Section 2.1.3 of the *UN Recommendations on the Transport of Dangerous Goods*, and Part I of the *UN Manual of Tests and Criteria*. Articles and substances classified in Class 1 must be assigned to the appropriate division and compatibility group in accordance with the procedures and criteria in those documents.

3.1.5.4 Except for substances which are listed by their proper shipping name in Subsection 4.2–List of Dangerous Goods, goods must not be offered for transport as Class 1 until they have been subjected to the classification procedure prescribed in this Subsection. In addition, the classification procedure must be undertaken before a new product is offered for transport. In this context, a new product is one, which, in the opinion of the appropriate national authority, involves any of the following:

- (a) a new explosive substance, or a combination or a mixture of explosive substances, which is considered to be significantly different from other combinations or mixtures already classified;

- (b) a new design of an explosive article or an article containing a new explosive substance or a new combination or mixture of explosive substances;
- (c) a new design of package for an explosive article or substance including a new type of inner packaging.

Note:

The importance of this can be overlooked unless it is realized that a relatively minor change in an inner or outer packaging can be critical and can convert a lesser hazard into a mass explosion hazard.

3.1.5.5 The producer or other applicant for classification of the product must provide adequate information concerning the names and characteristics of all explosive substances in the product and must furnish the results of all relevant tests which have been done. It is assumed that all the explosive substances in a new article have been properly tested and then approved.

3.1.6 Nomenclature of Explosives

Definitions for the special terms used in describing explosives may be found in Appendix A.

3.1.7 Exclusion from Class 1

3.1.7.1 The appropriate national authority may exclude an article or substance from Class 1 by virtue of test results and the Class 1 definition.

3.1.7.2 Where a substance provisionally accepted into Class 1 is excluded from Class 1 by performing UN Test Series 6 on a specific type and size of package, this substance, when meeting the classification criteria or definition for another class or division, should be listed in Subsection 4.2–List of Dangerous Goods in that class or division with a special provision restricting it to the type and size of package tested.

3.1.7.3 Where a substance is assigned to Class 1 but is diluted to be excluded from Class 1 by UN Test Series 6, this diluted substance (hereafter referred to as a desensitized explosive) should be listed in Subsection 4.2–List of Dangerous Goods with an indication of the highest concentration which excluded it from Class 1 (see 3.3.1.6 and 3.4.1.3) and if applicable, the concentration below which it is no longer deemed subject to these Regulations. New solid desensitized explosives subject to these Regulations should be listed in Division 4.1 and new liquid desensitized explosives should be listed in

Class 3. When the desensitized explosive meets the criteria or definition for another class or division, the corresponding subsidiary hazards(s) should be assigned to it.

3.1.7.4 An article may be excluded from Class 1 when three unpackaged articles, each individually activated by its own means of initiation or ignition or external means to function in the designed mode, meet the following test criteria:

- (a) no external surface has a temperature of more than 65°C. A momentary spike in temperature up to 200°C is acceptable;
- (b) no rupture or fragmentation of the external casing or movement of the article or detached parts thereof of more than 1 m in any direction;

Note:

Where the integrity of the article may be affected in the event of an external fire, these criteria must be examined by a fire test. One such method is described in ISO 14451-2 using a heating rate of 80 K/min.

- (c) no audible report exceeding 135 dB(C) peak at a distance of 1 m;
- (d) no flash or flame capable of igniting a material such as a sheet of 80 ±10 g/m² paper in contact with the article; and
- (e) no production of smoke, fumes or dust in such quantities that the visibility in a 1 m³ chamber equipped with appropriately sized blow out panels is reduced more than 50% as measured by a calibrated light (lux) meter or radiometer located 1 m from a constant light source located at the midpoint on opposite walls. The general guidance on Optical Density Testing in ISO 5659-1 and the general guidance on the Photometric System described in Section 7.5 in ISO 5659-2 may be used or similar optical density measurement methods designed to accomplish the same purpose may also be employed. A suitable hood cover surrounding the back and sides of the light meter must be used to minimize effects of scattered or leaking light not emitted directly from the source.

Notes:

1. If during the tests addressing criteria (a), (b), (c) and (d) no or very little smoke is observed the test described in (e) may be waived.
2. The appropriate national authority may require testing in packaged form if it is determined that, as packaged for transport, the article may pose a greater hazard.

3.1.8 Classification Documentation

3.1.8.1 An appropriate national authority assigning an article or substance into Class 1 should confirm with the applicant that classification in writing.

3.1.8.2 An appropriate national authority classification document may be in any form and may consist of more than one page, provided pages are numbered consecutively. The document should have a unique reference.

3.1.8.3 The information provided must be easy to identify, legible and durable.

3.1.8.4 Examples of the information that may be provided in the classification documents are as follows:

- (a) the name of the appropriate national authority and the provisions in national legislation under which it is granted its authority;
- (b) the modal or national regulations for which the classification document is applicable;
- (c) confirmation that the classification has been approved, made or agreed in accordance with the UN Model Regulations or these Regulations;
- (d) the name and address of the person in law to which the classification has been assigned and any company registration which uniquely identifies a company or other body corporate under national legislation;
- (e) the name under which the explosives will be placed onto the market or otherwise supplied for transport;
- (f) the proper shipping name, UN number, class, hazard division and corresponding compatibility group of the explosives;
- (g) where appropriate, the maximum net explosive mass of the package or article;
- (h) the name, signature, stamp, seal or other identification of the person authorised by the appropriate national authority to issue the classification document is clearly visible;
- (i) where safety in transport or the hazard division is assessed as being dependent upon the packaging, the packaging mark or a description of the permitted:
 - inner packagings;
 - intermediate packagings;
 - outer packagings.
- (j) the classification document states the part number, stock number or other identifying reference under which the explosives will be placed onto the market or otherwise supplied for transport;
- (k) the name and address of the person in law who manufactured the explosives and any company registration which uniquely identifies a company or other body corporate under national legislation;
- (l) any additional information regarding the applicable packing instruction and special packing provisions where appropriate;
- (m) the basis for assigning the classification, i.e. whether on the basis of test results, default for fireworks, analogy with classified explosive, by definition from the List of Dangerous Goods, etc.;
- (n) any special conditions or limitations that the appropriate national authority has identified as relevant to the safety for transport of the explosives, the communication of the hazard and international transport;
- (o) the expiry date of the classification document is given where the appropriate national authority considers one to be appropriate.

3.2 Class 2—Gases

STATE VARIATIONS: BHG-02, GHG-02, USG-02

3.2.1 Definition

3.2.1.1 A gas is a substance which:

- (a) at 50°C has a vapour pressure greater than 300 kPa (3.0 bar); or
- (b) is completely gaseous at 20°C at a standard pressure of 101.3 kPa (1.01 bar).

3.2.1.2 The transport condition of a gas is described according to its physical state as:

- (a) **Compressed gas**—a gas which, when packaged under pressure for transport, is entirely gaseous at -50°C; this category includes all gases with a critical temperature less than or equal to -50°C;
- (b) **Liquefied gas**—a gas which, when packaged under pressure for transport, is partially liquid at temperatures above -50°C. A distinction is made between:
 - **High pressure liquefied gas**—a gas with a critical temperature between -50°C and +65°C; and
 - **Low pressure liquefied gas**—a gas with a critical temperature above +65°C;
- (c) **Refrigerated liquefied gas**—a gas which, when packaged for transport, is made partially liquid because of its low temperature; or
- (d) **Dissolved gas**—a gas which, when packaged under pressure for transport, is dissolved in a liquid phase solvent.
- (e) **Adsorbed gas**—a gas which when packaged for transport is adsorbed onto a solid porous material resulting in an internal receptacle pressure of less than 101.3 kPa at 20°C and less than 300 kPa at 50°C.

3.2.1.3 This class comprises compressed gases; liquefied gases; dissolved gases; refrigerated liquefied gases; mixtures of one or more gases with one or more vapours of substances of other classes; articles charged with a gas, aerosols and chemicals under pressure. (For aerosols see 3.2.5).

Note:

"Cryogenic liquid" means the same as "refrigerated liquefied gas".

3.2.1.4 Pressures of all kinds relating to receptacles (such as test pressure, internal pressure, safety-valve opening pressure) are always indicated in gauge pressure (pressure in excess of atmospheric pressure); however, the vapour pressure of substances is always expressed in absolute pressure.

3.2.2 Divisions

3.2.2.0 Classification

Substances of Class 2 are assigned to one of three divisions based on the primary hazard of the gas during transport.

Note:

Aerosols (UN 1950), Receptacles, small, containing gas (UN 2037) and Gas cartridges (UN 2037) must be regarded as being in Division 2.1 when the criteria in 3.2.5.2.1 are met. For chemicals under pressure, UN 3500 to UN 3505, see Special Provision A187.

3.2.2.1 Division 2.1 Flammable Gas

Gases which at 20°C and a standard pressure of 101.3 kPa (1.01 bar):

- (a) are ignitable when in a mixture of 13% or less by volume with air; or
- (b) have a flammable range with air of at least 12 percentage points regardless of the lower flammable limit. Flammability must be determined by tests or by calculation in accordance with methods adopted by ISO (see ISO Standard 10156:2017). Where insufficient data are available to use these methods, tests by a comparable method recognized by the appropriate national authority must be used.

3.2.2.2 Division 2.2 Non-flammable, Non-toxic Gas

Gases which:

- (a) are asphyxiant—gases which dilute or replace the oxygen normally in the atmosphere; or
- (b) are oxidizing—gases which may, generally by providing oxygen, cause or contribute to the combustion of other material more than air does; or
- (c) do not come under the other divisions.

Note:

In 3.2.2.2(b) "gases which cause or contribute to the combustion of other material more than air does" means pure gases or gas mixtures with an oxidizing power greater than 23.5% as determined by a method specified in ISO 10156:2017.

3.2.2.3 Division 2.3 Toxic Gas

Gases which:

- (a) are known to be so toxic or corrosive to humans as to pose a hazard to health; or
- (b) are presumed to be toxic or corrosive to humans because they have an LC₅₀ value equal to or less than 5,000 mL/m³ (ppm) when tested in accordance with 3.6.1.5.3.

Note:

Gases meeting the above criteria owing to their corrosivity are to be classified as toxic with a subsidiary corrosive hazard.

3.2.2.4 Exceptions

3.2.2.4.1 Gases of Division 2.2, are not subject to these Regulations if they are transported at a pressure less than 200 kPa at 20°C and are not liquefied or refrigerated liquefied gases.

3.2.2.4.2 Gases of Division 2.2 are not subject to these Regulations when contained in the following:

- (a) foodstuffs, including carbonated beverages (except UN 1950);
- (b) balls intended for use in sports; or
- (c) tyres which meet the provisions of Special Provision A59.

Note:

This exception does not apply to lamps. For lamps see 1.2.11.

3.2.3 Mixtures of Gases

3.2.3.1 For the classification of gas mixtures into one of the three divisions (including vapours of substance from other classes) the following principles must be used:

3.2.3.1.1 Flammability must be determined by tests or by calculation in accordance with methods adopted by ISO (see ISO Standard 10156:2017). Where insufficient data are available to use these methods, tests by a comparable method recognized by the appropriate national authority may be used.

3.2.3.1.2 The level of toxicity is determined by either tests in accordance with 3.6.1.5.3 or a calculation method using the following formula:

$$LC_{50} \text{ Toxic (mixture)} = \frac{1}{\sum_{i=1}^n \frac{f_i}{T_i}}$$

where:

f_i = mole fraction of the i^{th} component substance of the mixture

T_i = toxicity index of the i^{th} component substance of the mixture (the T_c equals the LC_{50} value when available).

3.2.3.1.2.1 When LC_{50} values are unknown the toxicity index is determined by using the lowest LC_{50} value of substances of similar physiological and chemical effects, or through testing if this is the only practical possibility.

3.2.3.1.3 A gas mixture has a subsidiary hazard of corrosivity when the mixture is known by human experience to be destructive to the skin, eyes or mucous membranes or when the LC_{50} value of the mixture's corrosive components is equal to or less than 5,000 mL/m³ (ppm) when the LC_{50} is calculated by the formula:

$$LC_{50} \text{ Corrosive (mixture)} = \frac{1}{\sum_{i=1}^n \frac{f_c}{T_c}}$$

where:

f_c = mole fraction of the i^{th} corrosive component substance of the mixture

T_c = toxicity index of the i^{th} corrosive component substance of the mixture (the T_c equals the LC_{50} value when available).

3.2.3.1.4 Oxidizing ability is determined either by tests or by calculation methods adopted by the International Standards Organization (ISO) (see the Note under 3.2.2.2, ISO 10156:2017).

3.2.4 Hazard Precedence

Gases and gas mixtures with hazards associated with more than one division take the following precedence:

- Division 2.3 takes precedence over all other divisions;
- Division 2.1 takes precedence over Division 2.2.

3.2.5 Aerosols or Aerosol Dispensers

3.2.5.1 Definition

STATE VARIATION: USG-06

Class 2 also includes "aerosols". For the purpose of these Regulations an aerosol or aerosol dispenser means any non-refillable receptacle made of metal, glass or plastic and containing a gas compressed, liquefied or dissolved under pressure, with or without a liquid, paste or powder and fitted with a release device allowing the contents to be ejected as solid or liquid particles in suspension in a gas, as a foam, paste or powder, or in a liquid or gaseous state.

3.2.5.2 Classification

For aerosols, the division of Class 2 and the subsidiary hazards depend on the nature of the contents of the aerosol dispenser. The following provisions apply:

3.2.5.2.1 An aerosol must be assigned to Division 2.1 if the contents include 85% by weight or more of flammable components and the chemical heat of combustion is 30 kJ/g or more.

3.2.5.2.2 An aerosol must be assigned to Division 2.2 if the contents contain 1% by weight or less of flammable components and the heat of combustion is less than 20 kJ/g.

3.2.5.2.3 Aerosols not meeting the provisions of 3.2.5.2.1 or 3.2.5.2.2 must be classified in accordance with the tests described in the *UN Manual of Tests and Criteria*, Part III, section 31. Extremely flammable and flammable aerosols must be classified in Division 2.1; non-flammable in Division 2.2.

3.2.5.2.4 Gases of Division 2.3 must not be used as a propellant in an aerosol dispenser.

3.2.5.2.5 Where the contents other than the propellant of aerosol dispensers to be ejected are classified as Division 6.1, Packing Groups II or III or Class 8, Packing Groups II or III, the aerosol must have a subsidiary hazard of Division 6.1 or Class 8.

3.2.5.3 Flammable Aerosols

Flammable components are flammable liquids, flammable solids or flammable gases and gas mixtures as defined in Notes 1 to 3 of subsections 31.1.3 of Part III of the *UN Manual of Tests and Criteria*. This designation does not cover pyrophoric, self-heating or water-reactive substances. The chemical heat of combustion must be determined by one of the following methods ASTM D 240, ISO/FDIS 13943: 1999 (E/F) 86.1 to 86.3 or NFPA 30B.

3.2.5.4 Toxic Aerosols

Gases of Division 2.3 contained in an aerosol are forbidden from transport.

3.2.5.5 Aerosols in Packing Group I and II

3.2.5.5.1 Aerosols with contents meeting the criteria for Packing Group I for toxicity or corrosivity are forbidden from transport.

3.2.5.5.2 Aerosols with contents meeting the criteria for Packing Group II for toxicity or corrosivity are forbidden from transport unless shipped under the provisions of an exemption.

3.2.6 Gases Forbidden for Transport

Chemically unstable gases of Class 2 are forbidden for transport unless the necessary precautions have been taken to prevent the possibility of a dangerous decomposition or polymerization under normal conditions of transport. For the precautions necessary to prevent polymerization, see Special Provision A209. To this end particular care must be taken to ensure that receptacles do not contain any substances liable to promote these reactions.

3.3 Class 3—Flammable Liquids

3.3.1 Definition

3.3.1.1 Class 3 includes the following substances:

- (a) Flammable liquids (see 3.3.1.2 to 3.3.1.5); and
- (b) Liquid desensitized explosives (see 3.3.1.6).

3.3.1.2 This class has no subdivisions. It comprises liquids or mixtures of liquids or liquids containing solids in solution or in suspension (for example paints, varnishes, lacquers, etc., but not including substances otherwise classified on account of their dangerous characteristics) which give off a flammable vapour at temperatures of not more than 60°C closed-cup test or not more than 65.6°C open-cup test normally referred to as the flash point.

3.3.1.3 Liquids described in 3.3.1.2 with a flash point exceeding 35°C which do not sustain combustion need not be considered as flammable liquids for the purposes of these Regulations, if:

- (a) they have passed a suitable test for combustibility (see Sustained Combustibility Test prescribed in the *UN Manual of Tests and Criteria, Part III, subsection 32.5.2*); or

△ (b) their fire point according to ISO 2592:2000 is greater than 100°C; or

- (c) they are miscible solutions with a water content of more than 90% by weight.

3.3.1.4 Notwithstanding 3.3.1.2 and 3.3.1.3, liquids offered for transport at temperatures at or above their flash point are considered as flammable liquids.

3.3.1.5 Substances that are transported or offered for transport at elevated temperatures in a liquid state and which give off a flammable vapour at a temperature at or below the maximum transport temperature (i.e. the maximum temperature likely to be encountered by the substance in transport) are also considered to be flammable liquids.

3.3.1.6 Liquid desensitized explosives are explosive substances which are dissolved or suspended in water or other liquid substances, to form a homogeneous liquid mixture to suppress their explosive properties (see 3.1.7.3). Entries in Subsection 4.2—List of Dangerous Goods for liquid desensitized explosives are: UN 1204; UN 2059; UN 3064; UN 3343, UN 3357 and UN 3379.

Notes:

1. The word "flammable" has the same meaning as "inflammable."
2. The flash point of a flammable liquid may be altered by the presence of an impurity. The substances listed in Class 3 in Subsection 4.2—List of Dangerous Goods must generally be regarded as chemically pure. Since commercial products may contain added substances or impurities, flash points may vary and this may have an effect on classification or determination of the packing group for the product. In the event of doubt regarding the classification or packing group of a substance, the flash point of the substance must be determined experimentally.
3. Since the results of open-cup tests and closed-cup tests are not strictly comparable and even individual results by the same tests are often variable, regulations varying from the above figures to make allowances for such differences would be within the spirit of this definition.

3.3.2 Packing Group Criteria

3.3.2.1 Flammable liquids are assigned to packing groups according to the flash point and the boiling point of the liquid.

3.3.2.2 Use Table 3.3.A to determine the packing group of a liquid that presents a hazard only due to flammability.

TABLE 3.3.A
Class 3—Packing Group Assignment (3.3.2.2)

Packing Group	Flash Point (closed-cup)	Initial Boiling Point
I	—	≤ 35°C
II	< 23°C	> 35°C
III	≥ 23°C but ≤ 60°C	

3.3.2.3 For a liquid possessing an additional hazard(s), the packing group determined from Table 3.3.A and the packing group based on the severity of the additional hazard(s) must be considered. In such cases, the precedence of hazard characteristics appearing in Table 3.10.A should be used to determine the correct classification and packing group of the liquid.

3.3.3 Viscous Substances

3.3.3.1 Criteria for Inclusion in Packing Group III

3.3.3.1.1 Viscous flammable liquids such as paints, enamels, lacquers, varnishes, adhesives and polishes having a flash point of less than 23°C may be assigned to Packing Group III in conformity with the procedures prescribed in the UN *Manual of Tests and Criteria* Part III, subsection 32.3 provided that:

- (a) the viscosity* and flash point are in accordance with Table 3.3.B;

* *Viscosity determination: Where the substance concerned is non-Newtonian, or where a flow cup method of viscosity determination is otherwise unsuitable, a variable shear-rate viscometer must be used to determine the dynamic viscosity coefficient of the substance, at 23°C, at a number of shear rates. The values obtained are plotted against shear rate and then extrapolated to zero shear rate. The dynamic viscosity thus obtained, divided by the density, gives the apparent kinematic viscosity at near-zero shear rate.*

TABLE 3.3.B
Viscous Substances Packing Group III
Criteria (3.3.3.1.1(a))

Kinematic viscosity (extrapolated) ν (at near-zero shear rate) mm ² /s at 23°C	Flow time (t) in seconds	Jet diameter (mm)	Flash point (°C, closed cup)
20 < ν ≤ 80	20 < t ≤ 60	4	above 17
80 < ν ≤ 135	60 < t ≤ 100	4	above 10
135 < ν ≤ 220	20 < t ≤ 32	6	above 5
220 < ν ≤ 300	32 < t ≤ 44	6	above -1
300 < ν ≤ 700	44 < t ≤ 100	6	above -5
700 < ν	100 < t	6	-5 and below

- (b) less than 3% of the clear solvent layer separates in the solvent separation test;
- (c) the mixture or any separated solvent does not meet the criteria for Division 6.1 or Class 8;

- (d) the net quantity per package does not exceed 30 L for passenger aircraft or 100 L for cargo aircraft. Transport in accordance with this provision must be noted on the Shipper's Declaration.

Note:

When transported in accordance with this provision and packed with other dangerous goods in one outer package, the maximum quantity per package used to calculate the Q value for substances assigned to PG III in accordance with the provision must be the net quantity specified above.

3.3.3.1.2 Substances classified as flammable liquids due to their being transported or offered for transport at elevated temperature are included in Packing Group III.

Note:

Elevated temperature liquids are normally forbidden from transportation by air.

3.3.4 Determination of Flash Point

The following methods for determining the flash point of flammable liquids may be used:

International Standards

- ISO 1516
- ISO 1523
- ISO 2719
- ISO 3679
- ISO 3680
- ISO 13736

National Standards

- (a) Association française de normalisation (AFNOR)
11 avenue Francis de Pressencé
93571 Saint-Denis la Plaine
Cedex, France
Website: www.afnor.fr
Website: www.boutique-normes.afnor.org
- Petroleum products: NF-M-07-011, NF-M-07-019, NF-M-07-036
 - Paints, pigments and varnishes: NF-T-30-050
 - Black products (tar, etc.): NF-T-66-009
- (b) Deutsches Institut für Normung
Burggrafenstrasse 6
D-10787
Berlin
- Standard DIN 51755 (flash points below 65°C)
- (c) State Committee of the Council of Ministers for Standardization
Leninsky Prospect 9
113813, GSP
Moscow—49
- GOST 12.1.044-84
- (d) American Society for Testing Materials
100 Barr Harbor Drive, PO Box C700
West Conshohocken
Pennsylvania
19428-2959
- ASTM D 3828-93, Standard Test Methods for Flash Point by Small Scale Closed Tester

- ASTM D 56-93, Standard Test Method for Flash Point by Tag Closed Tester
- ASTM 3278-96, Standard Test Methods for Flash Point of Liquids by Setaflash Closed-Cup Apparatus
- ASTM 0093-96, Standard Test Methods for Flash Point by Pensky-martens Closed-Cup Tester

3.3.5 Determination of Initial Boiling Point

The following methods for determining the initial boiling point of flammable liquids may be used:

International standards

- ISO 3924
- ISO 4626
- ISO 3405

National standards

American Society for Testing Materials
100 Barr Harbor Drive, PO Box C700
West Conshohocken
Pennsylvania
19428-2959

- ASTM D86-07a, Standard Test Method for Distillation of Petroleum Products at Atmospheric Pressure
- ASTM D1078-05, Standard Test Method for Distillation Range of Volatile Organic Liquids

Further acceptable methods

- Method A.2 as described in Part A of the Annex to Commission Regulation (EC) No 440/2008

3.3.6 Substances Forbidden for Transport

Chemically unstable substances of Class 3 are forbidden for transport unless the necessary precautions have been taken to prevent the possibility of a dangerous decomposition or polymerization under normal conditions of transport. For the precautions necessary to prevent polymerization, see Special Provision A209. To this end particular care must be taken to ensure that receptacles and tanks do not contain any substances liable to promote these reactions.

3.4 Class 4—Flammable Solids; Substances Liable to Spontaneous Combustion; Substances which, in Contact with Water, Emit Flammable Gases

3.4.0 General

3.4.0.1 Class 4 is divided into three divisions as follows:

- Division 4.1 Flammable solids; Self-Reactive Substances; Polymerizing Substances; and Solid Desensitized Explosives.

- Division 4.2 Substances liable to spontaneous combustion.
- Division 4.3 Substances which, in contact with water, emit flammable gases (Dangerous when wet).

Notes:

1. Where the term "water-reactive" is used in these Regulations, it refers to a substance, which in contact with water, emits flammable gas.
2. Because of the different properties exhibited by the dangerous goods within Divisions 4.1 and 4.2, it is impracticable to establish a single criterion for classification in either of these divisions. Tests and criteria for assignment to the three divisions of Class 4 are given in 3.4.1, 3.4.2 and 3.4.3 and also in the UN Manual of Tests and Criteria, Part III, Section 33.
3. Since organometallic substances can be classified in Divisions 4.2 or 4.3 with additional subsidiary hazards, depending on their properties, a specific classification flow chart for these substances is given in Figure 2.4.2 of the UN Recommendations on the Transport of Dangerous Goods.

3.4.0.2 As referenced in this subsection, test methods and criteria, with advice on application of the tests, are given in the current edition of the *UN Manual of Tests and Criteria*, for the classification of the following types of substances of Class 4:

- (a) Flammable solids (Division 4.1);
- (b) Self-reactive substances (Division 4.1);
- (c) Polymerizing substances (Division 4.1);
- (d) Pyrophoric solids (Division 4.2);
- (e) Pyrophoric liquids (Division 4.2);
- (f) Self-heating substances (Division 4.2); and
- (g) Substances which, in contact with water, emit flammable gases (Division 4.3).

Test methods and criteria for self-reactive substances and polymerizing substances are given in the *UN Manual of Test and Criteria, Part II* and test methods and criteria for other types of substances of Class 4 are given in the *UN Manual of Tests and Criteria, Part III, section 33*.

3.4.1 Division 4.1—Flammable Solids; Self-Reactive Substances; Polymerizing Substances; and Solid Desensitized Explosives

Solids which, under conditions encountered in transport, are readily combustible or may cause or contribute to fire through friction; self-reactive substances and polymerizing substances which are liable to undergo a strongly exothermic reaction; desensitized explosives which may explode if not diluted sufficiently. Division 4.1 contains:

- flammable solids (3.4.1.1);
- self-reactive substances (3.4.1.2);
- solid desensitized explosives (3.4.1.3);
- polymerizing substances (3.4.1.4).

3.4.1.1 Flammable Solids

3.4.1.1.1 Properties

Flammable solids are readily combustible solids and solids which may cause fire through friction. Readily combustible solids are powdered, granular or pasty substances which are dangerous if they can be easily ignited by brief contact with an ignition source, such as a burning match and if the flame spreads rapidly. The danger may not only come from the fire but also from toxic combustion products. Metal powders are especially dangerous because of the difficulty of extinguishing a fire since normal extinguishing agents such as carbon dioxide or water can increase the hazard.

3.4.1.1.2 Classification of Flammable Solids

3.4.1.1.2.1 Powdered, Granular or Pasty Substances Other Than Metal Powders

Powdered, granular or pasty substances must be classified as readily combustible solids in Division 4.1 when the time of burning of one or more of the test runs, performed in accordance with the test method and criteria in the *UN Manual of Tests and Criteria*, Part III, subsection 33.2.1 is less than 45 seconds or the rate of burning is more than 2.2 mm per second.

3.4.1.1.2.2 Metal Powders

Powders of metal or metal alloys must be classified in Division 4.1 when they can be ignited and the reaction spreads over the whole length of the sample in 10 minutes or less.

3.4.1.1.2.3 Solids which may Cause Fire through Friction

Solids which may cause or contribute to fire through friction must be classified in Division 4.1 by analogy with existing entries (e.g. matches) until definitive criteria are established.

3.4.1.1.3 Assignment of Packing Groups

3.4.1.1.3.1 Other Than Metal Powders

Readily combustible powdered, granular or pasty substances, other than metal powders are allocated packing groups as follows:

- Packing Group II if the burning time measured in the test is less than 45 seconds and the flame passes the wetted zone; or
- Packing Group III if the burning time measured in the test is less than 45 seconds and the wetted zone stops the flame propagation for at least four minutes.

3.4.1.1.3.2 Metal Powders

Metal powders or metal alloys must be allocated packing groups as follows:

- Packing Group II if the zone of reaction observed during the test spreads over the whole length of the sample in 5 minutes or less; or
- Packing Group III if the zone of reaction observed during the test spreads over the whole length of the

sample in more than 5 minutes but not more than 10 minutes.

3.4.1.2 Self-Reactive Substances

3.4.1.2.1 Definition

Self-reactive substances of Division 4.1 are thermally unstable substances liable to undergo a strongly exothermic decomposition even without the participation of oxygen (air). The following substances must not be considered to be self-reactive substances of Division 4.1:

- explosives according to the criteria of Class 1;
- oxidizing substances according to the classification procedure for Division 5.1 (see 3.5.1.2) except that mixtures of oxidizing substances which contain 5.0% or more of combustible organic substances must be subjected to the classification procedure defined in 3.4.1.2.2.2 to 3.4.1.2.2.4;
- organic peroxides according to the criteria of Division 5.2 (see 3.5.2);
- substances where their heat of decomposition is less than 300 J/g; or
- substances where their self-accelerating decomposition temperature is greater than 75°C for a 50 kg package.

Note:

The heat of decomposition can be determined by using any internationally recognized method, e.g. differential scanning calorimetry and adiabatic calorimetry.

3.4.1.2.2 Criteria for Classification

3.4.1.2.2.1 Any substance, which shows the properties of a self-reactive substance, must be classified as such, even if this substance gives a positive test result for inclusion in Division 4.2 according to 3.4.2.3.3.

3.4.1.2.2.2 Mixtures of oxidizing substances meeting the criteria of Division 5.1 which contain 5.0% or more of combustible organic substances which do not meet the criteria mentioned in 3.4.1.2.1 (a), (c), (d) or (e) must be subjected to the self-reactive substance classification procedure.

3.4.1.2.2.3 A mixture showing the properties of a self-reactive substance type B to F must be classified as a self-reactive substance of Division 4.1.

3.4.1.2.2.4 A mixture showing the properties of a self-reactive substance type G according to the principle of 2.4.2.3.3.2 (g) of the *UN Recommendations* must be considered for classification as a substance of Division 5.1 (see 3.5.1.2).

3.4.1.2.3 Properties

The decomposition of self-reactive substances can be initiated by heat, contact with catalytic impurities, (e.g. acids, heavy-metal compounds, bases), friction or impact. The rate of decomposition increases with temperature and varies with the substance. Decomposition, particularly if no ignition occurs, may result in the evolution of toxic gases or vapours. For certain self-reactive substances, the temperature must be controlled. Some self-reactive substances may decompose explosively,

particularly if confined; this characteristic may be modified by the addition of diluents or by the use of appropriate packagings. Some self-reactive substances burn vigorously. Self-reactive substances include some of the following types of compounds:

- aliphatic azo compounds (-C-N=N-C-);
- organic azides (-C-N₃);
- diazonium salts (-CN₂⁺Z⁻);
- N-nitroso compounds (-N=N=O); and
- aromatic sulphonylhydrazides (-SO₂-NH-NH₂).

This list is not exhaustive and substances with other reactive groups and some mixtures of substances may have similar properties.

3.4.1.2.4 Classification

- **3.4.1.2.4.1** Self-reactive substances are classified into seven types according to the degree of danger they present. The types of self-reactive substance range from type A, which is forbidden in any mode of transport, to type G, which is not subject to the provisions for self-reactive substances of Division 4.1. The classification of types B to F is directly related to the maximum quantity allowed in one packaging.

- △ **3.4.1.2.4.2** Self-reactive substances permitted for transport in packages are listed in Appendix C.1. For each permitted substance listed, the appropriate generic entry appearing in Subsection 4.2–List of Dangerous Goods (UN 3223 to UN 3240) is assigned and appropriate subsidiary hazards and remarks providing relevant information are given. Related substances specifically listed by name in Subsection 4.2–List of Dangerous Goods are UN 2956, UN 3242 and UN 3251. The generic entries specify:

- the self-reactive substance type (C to F);
- the physical state, i.e. liquid/solid;
- when temperature control is required.

3.4.1.2.4.3 Classification of self-reactive substances not listed in Appendix C.1 and assignment to a generic entry must be made by the appropriate authority of the State in which the dangerous goods were manufactured on the basis of a test report. Principles applying to the classification of such substances are provided in 2.4.2.3.3 of the UN Recommendations on the Transport of Dangerous Goods. The applicable classification procedures, test methods and criteria and an example of a suitable test report, are given in the current edition of the *UN Manual of Tests and Criteria, Part II*. The statement of approval must contain the classification and the relevant transport conditions.

3.4.1.2.5 Transport of Samples

Samples of self-reactive substances not listed in Appendix C.1, for which a complete set of test results is not available and which are to be transported for further testing or evaluation, may be assigned to one of the appropriate entries for “Self-reactive substances type C” provided the following conditions are met:

- (a) the available data indicate that the sample would be no more dangerous than self-reactive substances type B;

- (b) the sample is packed in a combination packaging consisting of a plastic inner packaging with a capacity not exceeding 0.5 L or 0.5 kg which is placed in a wooden box (4C1), plywood box (4D) or fibreboard box (4G) with a maximum net quantity per package not exceeding 1 L or 1 kg; and
- (c) the available data indicate that the control temperature, if any, is sufficiently low to prevent any dangerous decomposition and sufficiently high to prevent any dangerous phase separation.

3.4.1.2.6 Temperature Control Requirements

- △ With the exception of self-reactive solids of type B which are forbidden for transport by air under any circumstance, self-reactive substances, which require temperature control during transport are forbidden for transport by air unless exempted (see 1.2.6). Self-reactive substances must be subject to temperature control if their self-accelerating decomposition temperature (SADT) is less than or equal to 55°C. Test methods for determining the SADT are provided in the current edition of the *UN Manual of Tests and Criteria*. The test selected must be conducted in a manner which is representative of the package to be transported both in size and material of construction.

3.4.1.2.7 Desensitization of Self-Reactive Substances

3.4.1.2.7.1 In order to ensure safety in transport, self-reactive substances may be desensitized by the use of a diluent. When a diluent is used, the self-reactive substance must be tested with the diluent present in the concentration and form to be used in transport.

3.4.1.2.7.2 Diluents which may allow a self-reactive substance to concentrate to a dangerous extent in the event of leakage from the package must not be used.

3.4.1.2.7.3 The diluent must be compatible with the self-reactive substance. In this regard, compatible diluents are those solids or liquids which have no detrimental influence on the thermal stability and hazard type of the self-reactive substance.

Note:

During the course of transport, packages or unit load devices containing self-reactive substances of Division 4.1 must be protected from direct sunlight and all sources of heat and be placed in adequately ventilated areas.

3.4.1.3 Solid Desensitized Explosives

3.4.1.3.1 Solid desensitized explosives are explosive substances which are wetted with water or alcohols or are diluted with other substances to form a homogeneous solid mixture to suppress their explosive properties. Entries in Subsection 4.2–List of Dangerous Goods for solid desensitized explosives are UN 1310, 1320, 1321, 1322, 1336, 1337, 1344, 1347, 1348, 1349, 1354, 1355, 1356, 1357, 1517, 1571, 2555, 2556, 2557, 2852, 2907, 3317, 3319, 3344, 3364, 3365, 3366, 3367, 3368, 3369, 3370, 3376, 3380 and 3474.

3.4.1.3.2 Substances that:

- (a) have been provisionally accepted into Class 1 according to Test Series 1 and 2 but exempted from Test Series 6;
- (b) are not self-reactive substances of Division 4.1;
- (c) are not substances of Class 5:

are also assigned to Division 4.1. UN 2956, UN 3241, UN 3242 and UN 3251 are such entries.

3.4.1.4 Polymerizing Substances and Mixtures (Stabilized)

3.4.1.4.1 Definitions and Properties

3.4.1.4.1.1 Polymerizing substances are substances which, without stabilization, are liable to undergo a strongly exothermic reaction resulting in the formation of larger molecules or resulting in the formation of polymers under conditions normally encountered in transport. Such substances are considered to be polymerizing substances of Division 4.1 when:

- (a) their self-accelerating polymerization temperature (SAPT) is 75°C or less under the conditions (with or without chemical stabilization as offered for transport) and in the packaging in which the substance or mixture is to be transported;
- (b) they exhibit a heat of reaction of more than 300 J/g; and
- (c) they do not meet any other criteria for inclusion in Classes 1 to 8.

3.4.1.4.1.2 A mixture meeting the criteria of a polymerizing substance must be classified as a polymerizing substance of Division 4.1.

3.4.1.4.1.3 Polymerizing substances are subject to temperature control in transport if their self-accelerating polymerization temperature (SAPT) is 50°C or less in the packaging in which the substance is to be transported.

Note:

Substances meeting the criteria of a polymerizing substance and also for inclusion in Classes 1 to 8 are subject to the requirements of Special Provision A209.

3.4.2 Division 4.2—Substances Liable to Spontaneous Combustion

3.4.2.1 Definition

Division 4.2—Substances liable to spontaneous combustion. Substances, which are liable to spontaneous heating under normal conditions encountered in transport, or to heating up in contact with air and being then liable to catch fire.

The following types of substances are classified in Division 4.2:

- pyrophoric substances; and
- self-heating substances.

3.4.2.2 Properties

Self-heating of a substance is a process where the gradual reaction of that substance with oxygen (in the air) generates heat. If the rate of heat production exceeds the rate of heat loss, then the temperature of the substance will rise which, after an induction time, may lead to self-ignition and combustion. Two types of substances can be distinguished with spontaneous combustion properties.

3.4.2.2.1 Pyrophoric Substances

Pyrophoric substances (liquid or solid) including mixtures and solutions are substances which, even in small quantities, ignite within 5 minutes of coming in contact with air. These substances are the most liable to spontaneous combustion.

3.4.2.2.2 Self-Heating Substances

Self-heating substances are substances, which in contact with air without an additional energy supply are liable to self-heating. These substances will ignite only in large amounts (kilograms) and after long periods of time (hours or days).

3.4.2.3 Classification

3.4.2.3.1 Pyrophoric Solids

Solids are considered pyrophoric solids which must be classified in Division 4.2 if, in tests performed in accordance with the test methods and criteria of in the current edition of the *UN Manual of Tests and Criteria, Part III, subsection 33.4*, the sample ignites in one of the tests.

3.4.2.3.2 Pyrophoric Liquids

Liquids are considered pyrophoric liquids which must be classified in Division 4.2 if, in tests performed in accordance with the test methods and criteria of the current edition of the *UN Manual of Tests and Criteria, Part III, subsection 33.4*, the liquid ignites in the first part of the test, or if it ignites or chars the filter paper.

3.4.2.3.3 Self-Heating Substances

3.4.2.3.3.1 A substance must be classified as a self-heating substance of Division 4.2 if, in tests performed in accordance with the test method and criteria in the current edition of the *UN Manual of Tests and Criteria, Part III, subsection 33.4*:

- (a) a positive result is obtained in a test using a 25 mm sample cube at 140°C;
- (b) a positive result is obtained in a test using a 100 mm sample cube at 140°C and a negative result is obtained in a test using a 100 mm cube sample at 120°C and the substance is to be transported in packages with a volume of more than 3 m³;
- (c) a positive result is obtained in a test using a 100 mm sample cube at 140°C and a negative result is obtained in a test using a 100 mm cube sample at 100°C and the substance is to be transported in packages with a volume of more than 450 L; and

- (d) a positive result is obtained in a test using a 100 mm sample cube at 140°C and a positive result is obtained in the test using a 100 mm cube sample at 100°C.

3.4.2.3.3.2 Self-reactive substances which also give a positive result according to this test method must not be classified in Division 4.2 but in Division 4.1 (see 3.4.1.2.1).

3.4.2.3.3.3 A substance must not be classified in Division 4.2 if:

- (a) a negative result is obtained in the test using a 100 mm sample cube at 140°C;
- (b) a positive result is obtained in the test using a 100 mm sample cube at 140°C and a negative result is obtained in the test using a 25 mm cube sample at 140°C, a negative result is obtained in the test using a 100 mm cube sample at 120°C and the substance is to be transported in packagings with a volume of not more than 3 m³; or
- (c) a positive result is obtained in the test using a 100 mm sample cube at 140°C and a negative result is obtained in the test using a 25 mm cube sample at 140°C, a negative result is obtained in the test using a 100 mm cube sample at 100°C and the substance is to be transported in packagings with a volume not more than 450 L.

3.4.2.4 Packing Group Criteria

3.4.2.4.1 Pyrophoric Substances

Pyrophoric liquids and solids of Division 4.2 must be assigned to Packing Group I.

3.4.2.4.2 Self-Heating Substances

Self-heating substances must be allocated to packing groups in accordance with the following criteria:

3.4.2.4.2.1 Packing Group II

Packing Group II must be assigned if the substance gives a positive result in the test using a 25 mm cube sample at 140°C.

3.4.2.4.2.2 Packing Group III

Packing Group III must be assigned to the substance if:

- (a) a positive result is obtained in the test using a 100 mm sample cube at 140°C and a negative result is obtained in the test using a 25 mm cube sample at 140°C and the substance is to be transported in packagings with a volume of more than 3 m³;
- (b) a positive result is obtained in a test using a 100 mm sample cube at 140°C and a negative result is obtained in the test using a 25 mm cube sample at 140°C, a positive result is obtained in the test using a 100 mm cube sample at 120°C and the substance is to be transported in packagings with a volume of more than 450 L; or

- (c) a positive result is obtained in the test using a 100 mm sample cube at 140°C and a negative result is obtained in the test using a 25 mm cube sample at 140°C and a positive result is obtained in a test using a 100 mm cube sample at 100°C.

3.4.3 Division 4.3—Substances which, in Contact with Water, Emit Flammable Gases

3.4.3.1 Definition

Division 4.3—Substances, which, in contact with water, emit flammable gases (Dangerous when wet). Substances which, by interaction with water, are liable to become spontaneously flammable or to give off flammable gases in dangerous quantities.

Note:

Where the term "Water-reactive" is used in these Regulations, it refers to a substance, which in contact with water emits flammable gas.

3.4.3.2 Properties

Certain substances in contact with water, emit flammable gases which can form explosive mixtures with air. Such mixtures are easily ignited by all ordinary sources of ignition, for example, naked lights, sparking hand tools or unprotected lamps. The resulting blast wave and flames may endanger people and the environment. The test method in 3.4.3.3 must be used to determine whether the reaction of a substance with water leads to the development of a dangerous amount of gases which may be flammable. It must not be applied to pyrophoric substances.

3.4.3.3 Classification

Substances which, in contact with water, emit flammable gases must be classified in Division 4.3 if, in tests performed in accordance with the methods and criteria in the current edition of the *UN Manual of Tests and Criteria, Part III, subsection 33.5*:

- (a) spontaneous ignition takes place in any step of the test procedure; or
- (b) there is an evolution of a flammable gas at a rate greater than 1 L/kg of the substance per hour.

3.4.3.3.1 Principle of the Method

The test method can be applied to solid and liquid substances. It is not applicable to pyrophoric substances. The substance must be tested in its commercial form at ambient temperature (20°C) by bringing it into contact with water. If during any stage of the test the gas emitted ignites, then no further testing is necessary and the substance must be assigned to Division 4.3. If spontaneous ignition of the emitted gas does not occur, then the final stage of the test must be performed to determine the rate of emission of flammable gas. Whether a substance is a water-reactive substance of Division 4.3 and if so, whether Packing Group I, II or III should be assigned is decided on the basis of the test result.

3.4.3.4 Packing Group Criteria

Substances which, in contact with water, emit flammable gases must be allocated to packing groups as follows:

3.4.3.4.1 Packing Group I

If the substance reacts vigorously with water at ambient temperatures and generally demonstrates a tendency for the gas produced to ignite spontaneously, or which reacts readily with water at ambient temperatures such that the rate of evolution of flammable gas is equal to or greater than 10 L/kg of substance over any one minute period.

3.4.3.4.2 Packing Group II

If the substance reacts readily with water at ambient temperatures such that the maximum rate of evolution of flammable gas is equal to or greater than 20 L/kg of substance per hour and does not meet the criteria for Packing Group I.

3.4.3.4.3 Packing Group III

If the substance reacts slowly with water at ambient temperatures such that the maximum rate of evolution of flammable gas is greater than 1 L/kg of substance per hour and does not meet the criteria for Packing Groups I or II.

3.4.4 Organometallic Substances

3.4.4.1 Classification

Depending on their properties, organometallic substances may be classified in Divisions 4.2 or 4.3, as appropriate, in accordance with the flowchart scheme given in Figure 2.4.2 of the *UN Recommendations on the Transport of Dangerous Goods*.

3.5 Class 5—Oxidizing Substances and Organic Peroxides

3.5.0 General

Class 5 is divided into two divisions:

- Division 5.1 Oxidizing substances.
- Division 5.2 Organic peroxides.

Note:

Because of the different properties exhibited by dangerous goods within Divisions 5.1 and 5.2, it is impracticable to establish a single criterion for classification in either division. Tests and criteria for assignment to the two divisions of Class 5 are addressed in this subsection and in the UN Manual of Tests and Criteria.

3.5.1 Division 5.1—Oxidizing Substances

3.5.1.1 Definition

Oxidizing substances are substances which, in themselves are not necessarily combustible, but may generally cause or contribute to the combustion of other material by yielding oxygen. Such substances may be contained in an article.

3.5.1.2 Classification

3.5.1.2.1 Oxidizing substances are classified in Division 5.1 in accordance with the test methods, procedures and criteria in 3.5.1.3 and 3.5.1.4 and the current edition of the *UN Manual of Tests and Criteria*, Part III, Section 34. In the event of divergence between test results and known experience, the appropriate authority of the State in which the dangerous goods were manufactured must be consulted to establish the appropriate classification and packing group.

Note:

Where substances of this division are listed in 4.2, reclassification of those substances in accordance with these criteria need only be undertaken when this is necessary for safety.

3.5.1.2.2 By exception, solid ammonium nitrate-based fertilizers must be classified in accordance with the procedure as set out in the *UN Manual of Tests and Criteria*, Part III, Section 39.

3.5.1.2.3 For substances having other hazards, e.g. toxicity or corrosivity, the requirements of 3.10 must be met.

3.5.1.3 Solid Oxidizers

3.5.1.3.1 Classification Criteria

3.5.1.3.1.1 Tests are performed to measure the potential for a solid substance to increase the burning rate or burning intensity of a combustible substance when the two are thoroughly mixed. The procedure is given in the *UN Manual of Tests and Criteria*, Part III, subsection 34.4.1 (Test O.1) or alternatively, in subsection 34.4.3 (Test O.3). Tests are conducted on the substance to be evaluated mixed with dry fibrous cellulose in mixing ratios of 1:1 and 4:1, by mass, of sample to cellulose. The burning characteristics of the mixtures are compared:

- in the Test O.1, with the standard 3:7 mixture, by mass, of potassium bromate to cellulose. If the burning time is equal to or less than this standard mixture, the burning times should be compared with those from the Packing Group I or II reference standards, 3:2 and 2:3 ratios, by mass, of potassium bromate to cellulose respectively; or
- in the Test O.3, with the standard 1:2 mixture, by mass, of calcium peroxide to cellulose. If the burning rate is equal to or greater than this standard mixture, the burning rates must be compared with those from the Packing Group I or II reference standards 3:1 and

1:1 ratios, by mass, of calcium peroxide to cellulose, respectively.

3.5.1.3.1.2 The classification test results are assessed on the basis of:

- (a) the comparison of the mean burning time (for the Test O.1) or burning rate (for the Test O.3) with those of the reference mixtures; and
- (b) whether the mixture of substance and cellulose ignites and burns.

3.5.1.3.1.3 A solid substance is classified in Division 5.1 if the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits:

- (a) in the Test O.1, a mean burning time equal to or less than the mean burning time of a 3:7 mixture (by mass) of potassium bromate and cellulose, or
- (b) in the Test O.3, a mean burning rate equal to or greater than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose.

3.5.1.3.2 Assignment of Packing Groups

3.5.1.3.2.1 Test Procedures

Solid oxidizing substances are assigned to a packing group according to the test procedures in the current edition of the *UN Manual of Tests and Criteria*, Part III, Section 34.4.1 (Test O.1) or alternatively, in subsection 34.4.3 (Test O.3), in accordance with the following requirements:

(a) Test O.1:

1. Packing Group I: any substances which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibit a mean burning time less than the mean burning time of a 3:2 mixture, by mass, of potassium bromate and cellulose;
2. Packing Group II: any substances which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibit a mean burning time equal to or less than the mean burning time of a 2:3 mixture (by mass) of potassium bromate and cellulose and the criteria for Packing Group I are not met;
3. Packing Group III: any substances which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibit a mean burning time equal to or less than the mean burning time of a 3:7 mixture (by mass) of potassium bromate and cellulose and the criteria for Packing Groups I and II are not met.
4. Not Division 5.1: any substances which, in both the 4:1 and 1:1 sample-to-cellulose ratio (by mass) tested, do not ignite and burn, or exhibit mean burning times greater than that of a 3:7 mixture (by mass) of potassium bromate and cellulose.

(b) Test O.3:

1. Packing Group I: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate greater than the mean burning rate of a 3:1 mixture (by mass) of calcium peroxide and cellulose;

2. Packing Group II: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:1 mixture (by mass) of calcium peroxide and cellulose and the criteria for Packing Group I are not met;

3. Packing Group III: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose and the criteria for Packing Groups I and II are not met;

4. Not Division 5.1: any substance which, in both the 4:1 and 1:1 sample-to-cellulose ratio (by mass) tested, does not ignite and burn, or exhibits a mean burning rate less than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose.

3.5.1.3.2.2 Other Substances

For substances having other hazards, e.g. toxicity or corrosivity, the requirements of Subsection 3.10 must be met.

3.5.1.4 Liquid Oxidizers

3.5.1.4.1 Classification Criteria

3.5.1.4.1.1 A test is performed to determine the potential for a liquid substance to increase the burning rate or burning intensity of a combustible substance or for spontaneous ignition to occur when the two are thoroughly mixed. The procedure is given in the *UN Manual of Tests and Criteria*, Part III, Section 34.4.2 (Test O.2). It measures the pressure rise time during combustion. Whether a liquid is an oxidizing substance of Division 5.1 and if so, whether Packing Group I, II or III should be assigned, is decided on the basis of the test result (see also Precedence of Hazard Characteristics in Subsection 3.10).

3.5.1.4.1.2 The test results are assessed on the basis of:

- (a) whether the mixture of the substance and cellulose spontaneously ignites; and
- (b) the comparison of the mean time taken for the pressure to rise from 690 kPa to 2,070 kPa gauge with those of the reference substances.

3.5.1.4.1.3 A liquid substance is classified in Division 5.1 if the 1:1 mixture (by mass) of substance and cellulose tested, exhibits a mean pressure rise time less than or equal to the mean pressure rise of a 1:1 mixture (by mass) of 65% aqueous nitric acid and cellulose.

3.5.1.4.2 Assignment of Packing Groups

Liquid oxidizers are assigned to a packing group according to the test procedures in the current edition of the *UN Manual of Tests and Criteria*, Part III, Section 34, in accordance with the following requirements:

3.5.1.4.2.1 Packing Group I

Substances which, in the 1:1 mixture, by mass, of substance and cellulose tested, spontaneously ignite; or, the mean pressure rise time of a 1:1 mixture, by mass, of substance and cellulose is less than that of a 1:1 mixture, by mass, of 50% perchloric acid and cellulose.

3.5.1.4.2.2 Packing Group II

Substances which, in the 1:1 mixture, by mass, of substance and cellulose tested, exhibit a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture, by mass, of 40% aqueous sodium chlorate solution and cellulose; and the criteria for Packing Group I are not met.

3.5.1.4.2.3 Packing Group III

Substances which, in the 1:1 mixture, by mass, of substance and cellulose tested, exhibit a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture, by mass, of 65% aqueous nitric acid and cellulose; and the criteria for Packing Groups I and II are not met.

3.5.1.4.2.4 Not Division 5.1

Substances which, in the 1:1 mixture, by mass, of substance and cellulose tested, exhibit a pressure rise of less than 2,070 kPa gauge; or, exhibit a mean pressure rise time greater than the mean pressure rise time of a 1:1 mixture, by mass, of 65% aqueous nitric acid and cellulose.

3.5.1.4.2.5 Substances having other Hazards

For substances having other hazards, e.g. toxicity or corrosivity, the requirements of Subsection 3.10 must be met.

3.5.2 Division 5.2—Organic Peroxides

3.5.2.1 Definition

3.5.2.1.1 This division is made up of organic substances which contain the bivalent structure -O-O- and may be considered derivatives of hydrogen peroxide in which one or both of the hydrogen atoms have been replaced by organic radicals.

Note:

Hydrogen peroxide is made up of two hydrogen atoms and two oxygen atoms connected in a chain: H-O-O-H.

3.5.2.1.2 Organic peroxides are thermally unstable substances which may undergo exothermic, self-accelerating decomposition. In addition, they may have one or more of the following properties:

- be liable to explosive decomposition;
- burn rapidly;
- be sensitive to impact or friction;
- react dangerously with other substances;
- cause damage to the eyes.

3.5.2.2 Classification

3.5.2.2.1 Any organic peroxide must be considered for classification in Division 5.2 unless the organic peroxide formulation contains:

- not more than 1.0% available oxygen from the organic peroxides when containing not more than 1.0% hydrogen peroxide; or
- not more than 0.5% available oxygen from the organic peroxides when containing more than 1.0% but not more than 7.0% hydrogen peroxide.

Note:

The available oxygen content (percent) of an organic peroxide formulation is given by the formula:

$$O_A = 16 \times \sum \left(\frac{n_i \times C_i}{m_i} \right)$$

where:

O_A = available oxygen content (percent);

n_i = number of peroxygen groups per molecule of organic peroxide "i";

C_i = concentration (mass percent) of organic peroxide "i"; and

m_i = molecular mass of organic peroxide "i".

3.5.2.2.2 Organic peroxides are classified into seven types according to the degree of danger they present. The types of organic peroxide range from type A, which is forbidden in any mode of transport, to type G, which is not subject to the provisions for organic peroxides of Division 5.2. The classification of types B to F is directly related to the maximum quantity allowed in one packaging.

3.5.2.3 Type of Organic Peroxide

Organic peroxides permitted for transport are listed in Appendix C.2. For each permitted substance, Appendix C.2 assigns the appropriate generic entry appearing in Subsection 4.2—List of Dangerous Goods (UN 3103 to UN 3120) and provides relevant information. The generic entries specify:

- the organic peroxide type (B to F);
- the physical state (liquid or solid); and
- when temperature control is required (see 3.5.2.7).

Note:

Organic peroxides requiring temperature control during transport are forbidden for transport by air unless exempted (see 2.1.2) under the procedures outlined in 1.2.6.

3.5.2.3.1 Classification Approvals

Classification of organic peroxides not listed in Appendix C.2 and assignment to a generic entry must be made by the appropriate authority of the State in which the dangerous goods were manufactured on the basis of a test report. Principles applying to the classification of such substances are provided in 2.5.3.3 of the *UN Recommendations on the Transport of Dangerous Goods*. The applicable classification procedures, test

methods and criteria and an example of a suitable test report, are provided in the current edition of the *UN Manual of Test and Criteria, Part II*. The statement of approval must contain the classification and the relevant transport conditions.

3.5.2.3.2 Mixtures

Mixtures of the listed formulations may be classified as the same type of organic peroxide as that of the most dangerous component and be transported under the conditions of transport given for this type. However, as two stable components can form a thermally less stable mixture, the self-accelerating decomposition temperature (SADT) of the mixture must be determined and if necessary, temperature control applied as required by 3.5.2.7.

3.5.2.4 Properties

3.5.2.4.1 Organic peroxides are liable to exothermic decomposition, which can be started by heat, contact with impurities (e.g. acids, heavy metal compounds, amines), friction or impact. The rate of decomposition increases with temperature and varies with the peroxide formulation. Decomposition may result in the evolution of harmful or flammable gases or vapours. For certain organic peroxides, the temperature must be controlled during transport. Some organic peroxides decompose explosively, particularly if confined. This characteristic may be modified by the addition of diluents or by the use of appropriate packagings. Many organic peroxides burn vigorously.

3.5.2.4.2 Contact of organic peroxides with the eyes should be avoided. Certain organic peroxides will cause serious injury to the cornea even after brief contact, or will be corrosive to the skin.

3.5.2.4.3 During the course of transport, packages or unit load devices containing organic peroxides must be protected from direct sunlight and all sources of heat and be placed in adequately ventilated areas.

3.5.2.5 Desensitization

3.5.2.5.1 In many cases, organic peroxides are desensitized by the addition of organic liquids or solids, inorganic solids or water in order to ensure safety during transport and handling. Where a percentage of a substance is stipulated, this refers to the percentage by mass, rounded to the nearest whole number. In general, desensitization should be such that in the case of spillage or fire, the organic peroxide may not concentrate to a dangerous extent.

3.5.2.5.2 Unless otherwise stated for the individual organic peroxide formulation, the following definitions apply for diluents used for desensitization:

- (a) **Diluents type A** are organic liquids which are compatible with the organic peroxide and which have a boiling point of not less than 150°C. Type A diluents may be used for desensitizing all organic peroxides;
- (b) **Diluents type B** are organic liquids which are compatible with the organic peroxide and which have a boiling point of less than 150°C but not less than 60°C and a flashpoint of not less than 5°C. Type B diluents may be used for desensitization of all organ-

ic peroxides provided that the boiling point of the liquid is at least 60°C higher than the SADT in a 50 kg package.

3.5.2.5.3 Diluents, other than type A or type B, may be added to organic peroxide formulations as listed in Appendix C.2 provided that they are compatible. However, replacement of all or part of a type A or type B diluent by another diluent with differing properties requires that the organic peroxide formulation be reassessed in accordance with the normal acceptance procedure for Division 5.2.

3.5.2.5.4 Water may only be used for the desensitization of organic peroxides which are shown in Appendix C.2 or in the statement of approval according to 3.5.2.3.1 with the approval of the appropriate authority of the State of manufacture or when the organic peroxide formulation is specified as being with water or as a stable dispersion in water.

3.5.2.5.5 Organic and inorganic solids may be used for desensitization of organic peroxides provided that they are compatible.

3.5.2.5.6 Compatible liquids and solids are those which have no detrimental influence on the thermal stability and hazard type of the organic peroxide formulation.

3.5.2.6 Transport of Samples

Samples of new formulations of organic peroxides not listed in Appendix C.2 for which complete test data are not available and which are to be transported for further testing or evaluation may be assigned to one of the appropriate entries for "Organic peroxide, type C" provided that the following conditions are met:

- (a) the available data indicate that the sample would be no more dangerous than organic peroxides type B;
- (b) the sample is packed in a combination packaging consisting of a plastic inner packaging with a capacity not exceeding 0.5 L or 0.5 kg which is placed in a wooden box (4C1), plywood box (4D) or fibreboard box (4G) with a maximum net quantity per package not exceeding 1 L or 1 kg; and
- (c) the available data indicate that the control temperature, if any, is sufficiently low to prevent any dangerous decomposition and sufficiently high to prevent any dangerous phase separation.

3.5.2.7 Temperature Control Requirements

- △ **3.5.2.7.1** An organic peroxide must be regarded as possessing explosive properties when in laboratory testing the formulation is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement. With the exception of organic peroxides of type B which are forbidden for transport by air under any circumstance, organic peroxides requiring temperature control during transport are forbidden for transport by air unless shipped under an approval or exemption, as applicable (see 1.2.5 and 1.2.6).

3.5.2.7.2 The following organic peroxides must be subjected to temperature control during transport:

- (a) organic peroxides types B and C with an SADT $\leq 50^{\circ}\text{C}$;
- (b) organic peroxides type D showing a medium effect when heated under confinement with an SADT $\leq 50^{\circ}\text{C}$ or showing a low or no effect when heated under confinement with an SADT $\leq 45^{\circ}\text{C}$; and
- (c) organic peroxides types E and F with an SADT $\leq 45^{\circ}\text{C}$.

3.5.2.7.3 Test methods for determining the SADT are provided in the current edition of the *UN Manual of Tests and Criteria, Part III, Section 28*. The test selected must be conducted in a manner which is representative of the package to be transported.

3.5.2.7.4 Test methods for determining the flammability are provided in the *UN Manual of Tests and Criteria, Part III, subsection 32.4*.

3.6 Class 6—Toxic and Infectious Substances

3.6.0 General

Class 6 is divided into two divisions as follows:

- Division 6.1—Toxic substances.
- Division 6.2—Infectious substances.

3.6.1 Division 6.1—Toxic Substances

STATE VARIATION: USG-02

3.6.1.1 Definition

Division 6.1 Toxic substances are substances which are liable to cause death or injury or to harm human health if swallowed, inhaled or contacted by the skin.

Note:

Toxins from plant, animal or bacterial sources which do not contain any infectious substances or toxins that are contained in substances which are not infectious substances should be considered for classification in Division 6.1 and assignment to UN 3172.

3.6.1.2 Classification

To be classified as "Toxic" for purposes of these Regulations, a substance must meet the minimum criteria for Packing Group III shown in Tables 3.6.A and 3.6.B.

3.6.1.3 Packing Group Criteria

3.6.1.3.1 Toxic substances of Division 6.1, including pesticides, must be allocated to packing groups referred according to the degree of their toxic hazards in transport as follows:

- (a) **Packing Group I**—Substances and preparations presenting a very severe toxicity hazard;
- (b) **Packing Group II**—Substances and preparations presenting a serious toxicity hazard;
- (c) **Packing Group III**—Substances and preparations presenting a relatively low toxicity hazard.

3.6.1.3.2 In allocating the packing group, account must be taken of human experience in instances of accidental poisoning and of special properties possessed by any individual substance, such as liquid state, high volatility, any special likelihood of penetration and special biological effects.

3.6.1.3.3 In the absence of human experience the grouping must be based on the available data from animal experiments. Three possible routes of administration must be examined. These routes are exposure through:

- (a) oral ingestion;
- (b) dermal contact; and
- (c) inhalation of dusts, mists or vapours.

3.6.1.3.4 Appropriate animal tests for the various routes of exposure are described in 3.6.1.5.1 to 3.6.1.5.3. When a substance exhibits a different order of toxicity by two or more of these routes of administration, the highest degree of danger must be assigned.

3.6.1.3.5 The criteria to be applied for grouping a substance according to the toxicity it exhibits by all three routes of administration are presented in the following paragraphs.

3.6.1.3.6 The grouping for the oral and dermal routes as well as for inhalation of dusts and mists are as shown in Table 3.6.A.

3.6.1.4 Forbidden Toxic Substances

Liquids having a vapour inhalation toxicity of Packing Group I are forbidden on both passenger and cargo aircraft.

TABLE 3.6.A
Oral, Dermal and Dust/Mist Inhalation Hazards Division 6.1 Packing Group Criteria (3.6.1.3)

Packing Group	Oral Toxicity LD ₅₀ (mg/kg)	Dermal Toxicity LD ₅₀ (mg/kg)	Inhalation Toxicity by Dusts and Mists LC ₅₀ (mg/L)
I	≤ 5.0	≤ 50	≤ 0.2
II	> 5.0 but ≤ 50	> 50 but ≤ 200	> 0.2 but ≤ 2.0
III ^a	> 50 but ≤ 300	> 200 but ≤ 1,000	> 2.0 but ≤ 4.0

- ^a Tear gas substances must be included in Packing Group II even if their toxicity data correspond to Packing Group III values.

Notes:

- See 3.6.1.5 for explanation of LD₅₀ and LC₅₀.
- Substances meeting the criteria of Class 8 and with an inhalation toxicity of dusts and mists (LC₅₀) leading to Packing Group I are only accepted for an allocation to Division 6.1 if the toxicity through oral ingestion or dermal contact is at least in the range of Packing Group I or II. Otherwise an allocation to Class 8 is made when appropriate (see 3.8.2.3).

TABLE 3.6.B
Vapour Inhalation Hazard—Division 6.1 Packing Group Criteria (3.6.1.3)

Packing Group	Inhalation Hazard
I	LC ₅₀ ≤ 1,000 mL/m ³ and V ≥ 10 × LC ₅₀
II	LC ₅₀ ≤ 3,000 mL/m ³ and V ≥ LC ₅₀ and not meeting criteria for Packing Group I
III	LC ₅₀ ≤ 5,000 mL/m ³ and V ≥ 0.2 × LC ₅₀ and not meeting criteria for Packing Group I or II

Notes:

- V is that saturated vapour concentration in air of the substance in mL/m³ at 20°C and standard atmospheric pressure. If the vapour pressure at 20°C is known, V can be calculated as follows:

$$V = \frac{p}{P} 10^6 \text{ mL/m}^3$$

where:

p = vapour pressure at 20°C.

P = standard atmospheric pressure.

Note p and P must be the same, e.g.:

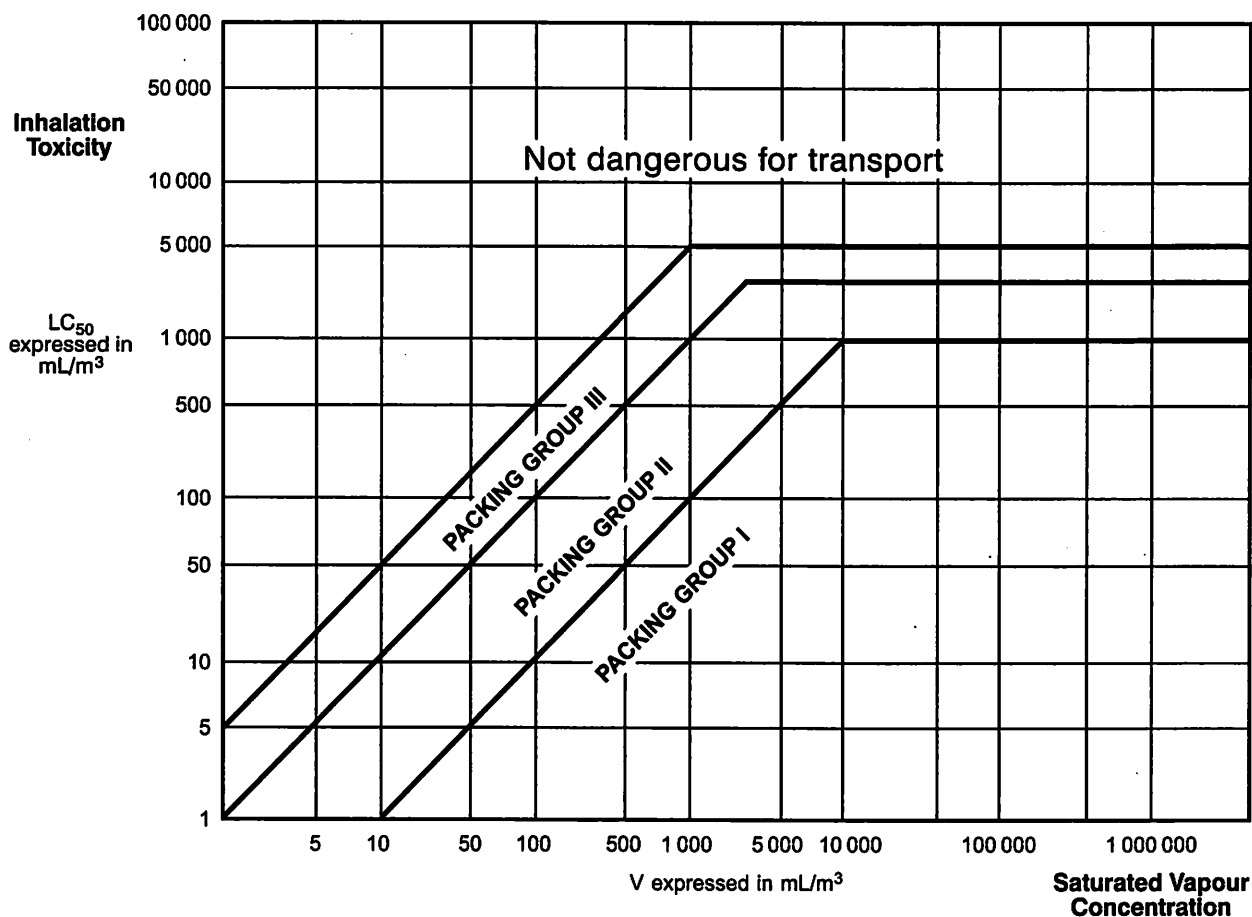
if p is in kPa, P = 100 kPa;

if p is in mBar, P = 1,000 mBar;

if p is in psi, P = 14.7 psi.

- Tear gas substances having toxicity data corresponding to Packing Group III values are nevertheless included in Packing Group II.
- Figure 3.6.C is a graphic representation of Table 3.6.B.

FIGURE 3.6.C
Vapour Inhalation Hazard Diagram—Division 6.1 Packing Group Criteria (3.6.B)



3.6.1.5 Tests—Division 6.1, Toxic Substances

The criteria listed in Tables 3.6.A and 3.6.B are derived from the test methods described in 3.6.1.5.1 to 3.6.1.5.3.

3.6.1.5.1 Oral Toxicity

The LD_{50} (median lethal dose) for acute oral toxicity is the statistically derived single dose of a substance which, when administered by mouth, can be expected to cause death within 14 days in 50% of young adult albino rats. The LD_{50} value is expressed in terms of mass of test substance per mass of test animal (mg/kg).

3.6.1.5.2 Dermal Toxicity

The LD_{50} for acute dermal toxicity is that dose of the substance which, when administered by continuous contact with the bare skin of an albino rabbit for 24 hours, is most likely to cause death within 14 days in half of the animals tested. The number of animals tested must be sufficient to give a statistically significant result and be in

conformity with good pharmacological practices. The result is expressed in mg/kg body mass.

3.6.1.5.3 Inhalation Toxicity

The LC_{50} for acute toxicity on inhalation is that concentration of dust, mist or vapour which, when administered by continuous inhalation for one hour to both male and female young adult albino rats, is most likely to cause death within 14 days in half of the animals tested. A solid substance should be tested if at least 10% (by mass) of its total mass is likely to be dust in a respirable range, e.g. the aerodynamic diameter of that particle-fraction is 10 μm or less. A liquid substance should be tested if a mist is likely to be generated in a leakage of the transport containment. Both for solid and liquid substances more than 90% (by mass) of a specimen prepared for inhalation toxicity should be in the respirable range as defined above. The result is expressed in mg/L of air for dusts and mists, or in mL/m^3 of air (parts per million) for vapours.

3.6.1.5.3.1 Inhalation Criteria for Dust and Mist

The criteria for inhalation toxicity of dusts and mists in 3.6.1.3 and Table 3.6.A are based on LC₅₀ data relating to 1 hour exposures and where such information is available it must be used. However, where only LC₅₀ data relating to 4 hour exposures to dusts and mists are available, such figures can be multiplied by four and the product substituted in the above criteria; i.e. LC₅₀ (4 h) × 4 is considered the equivalent of LC₅₀ (1 h).

3.6.1.5.3.2 Inhalation Criteria for Vapours

Liquids having toxic vapours must be assigned to packing groups shown in 3.6.1.3 and Table 3.6.B where V is that saturated vapour concentration in air of the substance in mL/m³ at 20°C and standard atmospheric pressure. The criteria for inhalation toxicity of vapours in 3.6.1.3 and Table 3.6.B are based on LC₅₀ data relating to 1 hour exposures and where such information is available it must be used. However, where only LC₅₀ data relating to 4 hour exposures to the vapours are available, such figures can be multiplied by two and the product substituted in the above criteria; i.e. LC₅₀ (4 h) × 2 is considered the equivalent of LC₅₀ (1 h). In Figure 3.6.C, the above criteria are expressed in graphical form, as an aid to easy classification. However, because of approximations inherent in the use of graphs, substances on or near packing group borderlines must be checked using numerical criteria.

3.6.1.6 Inhalation Criteria for Liquid Mixtures

Mixtures of liquids that are toxic by inhalation must be assigned to packing groups according to 3.6.1.6.1 or 3.6.1.6.2.

3.6.1.6.1 Calculations to Establish Packing Group

If LC₅₀ data is available for each of the toxic substances comprising a liquid mixture, the packing group may be determined as follows:

- (a) Estimate the LC₅₀ of the mixture using the formula:

$$LC_{50} \text{ (mixture)} = \frac{1}{\sum_{i=1}^n \frac{f_i}{LC_{50_i}}}$$

f_i = mole fraction of the i^{th} component substance of the liquid mixture; and

LC₅₀ = mean lethal concentration of the i^{th} component substance of the liquid mixture in mL/m³;

- (b) Estimate the volatility of each component substance of the liquid mixture using the following formula:

$$V_i = p_i \times \frac{10^6}{101.3} \text{ mL/m}^3$$

p_i = partial pressure of the i^{th} component substance in kPa at 20°C and one atmosphere pressure;

- (c) Calculate the ratio of the volatility to the LC₅₀ using the formula:

$$R = \sum_{i=1}^n \frac{V_i}{LC_{50_i}}$$

- (d) Using the calculated values for LC₅₀ (mixture) and for R, the packing group of the mixture is determined as follows:

- Packing Group I: $R \geq 10$ and LC₅₀ (mixture) $\leq 1,000 \text{ mL/m}^3$;
- Packing Group II: $R \geq 1$ and LC₅₀ (mixture) $\leq 3,000 \text{ mL/m}^3$ and not meeting criteria for Packing Group I;
- Packing Group III: $R \geq 0.2$ and LC₅₀ (mixture) $\leq 5,000 \text{ mL/m}^3$ and not meeting criteria for Packing Group I or II.

3.6.1.6.2 Tests to Establish Packing Group

In the absence of LC₅₀ data on the toxic component substances, the mixture may be assigned a packing group based on the following simplified threshold toxicity tests. When these threshold tests are used, the most restrictive packing group must be determined and used for transporting the mixture.

3.6.1.6.2.1 Packing Group I

A mixture is assigned to Packing Group I only if it meets both of the following criteria:

- (a) A sample of the liquid mixture is vaporized and diluted with air to create a test atmosphere of 1,000 mL/m³ vaporized mixture in air. Ten albino rats (five male and five female) are exposed to the test atmosphere for one hour and observed for 14 days. If five or more of the animals die within the 14 day observation period, the mixture is assumed to have an LC₅₀ equal to or less than 1,000 mL/m³; and
- (b) A sample of the vapour in equilibrium with the liquid mixture at 20°C is diluted with nine equal volumes of air to form a test atmosphere. Ten albino rats (five male and five female) are exposed to the test atmosphere for one hour and observed for 14 days. If five or more of the animals die within the 14 day observation period, the mixture is assumed to have a volatility equal to or greater than ten times the LC₅₀ of the mixture.

3.6.1.6.2.2 Packing Group II

A mixture is assigned to Packing Group II only if it meets both of the following criteria and the mixture does not meet the criteria for Packing Group I:

- (a) A sample of the liquid mixture is vaporized and diluted with air to create a test atmosphere of 3,000 mL/m³ vaporized mixture in air. Ten albino rats (five male and five female) are exposed to the test atmosphere for one hour and observed for 14 days. If five or more of the animals die within the 14 day observation period, the mixture is assumed to have an LC₅₀ equal to or less than 3,000 mL/m³; and

- (b) A sample of the vapour in equilibrium with the liquid mixture at 20°C is used to form a test atmosphere. Ten albino rats (five male and five female) are exposed to the test atmosphere for one hour and observed for 14 days. If five or more of the animals die within the 14 day observation period, the mixture is assumed to have a volatility equal to or greater than the LC₅₀ of the mixture.

3.6.1.6.2.3 Packing Group III

A mixture is assigned to Packing Group III only if it meets both of the following criteria and the mixture does not meet the criteria for Packing Group I or II:

- (a) A sample of the liquid mixture is vaporized and diluted with air to create a test atmosphere of 5,000 mL/m³ vaporized mixture in air. Ten albino rats (five male and five female) are exposed to the test atmosphere for one hour and observed for 14 days. If five or more of the animals die within the 14 day observation period, the mixture is assumed to have an LC₅₀ equal to or less than 5,000 mL/m³; and
- (b) The vapour pressure of the liquid mixture is measured and if the vapour pressure is equal to or greater than 1,000 mL/m³, the mixture is presumed to have a volatility equal to or greater than 0.2 or (1/5) the LC₅₀ of the mixture.

3.6.1.7 Methods for Determining Oral and Dermal Toxicity of Mixtures

When classifying and assigning the appropriate packing group to mixtures in Division 6.1, in accordance with the oral and dermal toxicity criteria in Table 3.6.A, it is necessary to determine the acute LD₅₀ of the mixture.

3.6.1.7.1 One Active Substance

If a mixture contains only one active substance and the LD₅₀ of that constituent is known, in the absence of reliable acute oral and dermal toxicity data on the actual mixture to be transported, the oral or dermal LD₅₀ may be obtained by the following method:

$$\text{LD}_{50} \text{ value of mixture} = \frac{\text{LD}_{50} \text{ value of active substance} \times 100}{\text{percentage of active substance by mass}}$$

3.6.1.7.2 More Than One Active Substance

If a mixture contains more than one active constituent, there are three possible approaches that may be used to determine the oral or dermal LD₅₀ of the mixture. The preferred method is to obtain reliable acute oral and dermal toxicity data on the actual mixture to be transported. If reliable, accurate data is not available, then either of the following methods in 3.6.1.7.2.1 or 3.6.1.7.2.2 may be performed.

3.6.1.7.2.1 Classify the formulation according to the most hazardous constituent of the mixture as if that constituent were present in the same concentration as the total concentration of all active constituents.

3.6.1.7.2.2 Apply the formula:

$$\frac{C_A}{T_A} + \frac{C_B}{T_B} + \dots + \frac{C_Z}{T_Z} = \frac{100}{T_m}$$

where:

C = the % concentration of constituent A, B ... Z in the mixture

T = the oral LD₅₀ values of constituent A, B ... Z

T_m = the oral LD₅₀ value of the mixture

Note:

This formula can also be used for dermal toxicities provided that this information is available on the same species for all constituents. The use of this formula does not take into account any potentiation or protective phenomena.

3.6.1.8 Pesticides

3.6.1.8.1 All active pesticide substances and their preparations for which the LC₅₀ and/or LD₅₀ values are known and which are classified in Division 6.1 must be classified under appropriate packing groups in accordance with the criteria given in 3.6.1.5. Substances and preparations which are characterized by subsidiary hazards must be classified according to the Precedence of Hazard Table 3.10.A with the assignment of appropriate packing groups.

3.6.1.8.2 If the oral or dermal LD₅₀ value for a pesticide preparation is not known, but the LD₅₀ value of its active substance(s) is known, the LD₅₀ value for the preparation may be obtained by applying the procedures in 3.6.1.7.

3.6.1.8.3 The proper shipping name used in the transport of the pesticide must be selected on the basis of the active ingredient, the physical state of the pesticide and any subsidiary hazards it may exhibit.

3.6.1.8.4 LD₅₀ toxicity data for a number of common pesticides may be obtained from the most current edition of the document the *WHO Recommended Classification of Pesticides by Hazard and Guidelines to Classification* available from the International Programme on Chemical Safety, World Health Organization (WHO), 1211 Geneva 27, Switzerland. While that document may be used as a source of LD₅₀ data for pesticides, its classification system should not be used for purposes of transport classification of, or assignment of packing groups to, pesticides, which must be in accordance with these Regulations.

3.6.1.9 Substances Forbidden for Transport

Chemically unstable substances of Division 6.1 are forbidden for transport unless the necessary precautions have been taken to prevent the possibility of a dangerous decomposition or polymerization under normal conditions of transport. For the precautions necessary to prevent polymerization, see Special Provision A209. To this end particular care must be taken to ensure that receptacles and tanks do not contain any substances liable to promote these reactions.

3.6.2 Division 6.2—Infectious Substances

STATE VARIATIONS: AUG-03, CAG-08, VUG-02

OPERATOR VARIATIONS: 4C-04, BZ-07, CM-05, FX-04, JJ-04, L7-04, LA-07, LP-04, LU-04, M3-04, P5-05, SN-03, UC-04, XL-04

3.6.2.1 Definitions

For the purposes of these Regulations:

3.6.2.1.1 Infectious substances are substances which are known or are reasonably expected to contain pathogens. Pathogens are defined as micro-organisms (including bacteria, viruses, parasites, fungi) and other agents such as prions, which can cause disease in humans or animals.



Note:

Toxins from plant, animal or bacterial sources which do not contain any infectious substances or toxins that are not contained in substances which are infectious substances should be considered for classification in Division 6.1 and assigned to UN 3172 or UN 3462.

3.6.2.1.2 Biological products are those products derived from living organisms which are manufactured and distributed in accordance with the requirements of appropriate national authorities, which may have special licensing requirements and are used either for prevention, treatment, or diagnosis of disease in humans or animals, or for development, experimental or investigational purposes related thereto. They include, but are not limited to, finished or unfinished products such as vaccines.

3.6.2.1.3 Cultures are the result of a process by which pathogens are intentionally propagated. This definition does not include patient specimens as defined below in 3.6.2.1.4.

3.6.2.1.4 Patient specimens are those collected directly from humans or animals, including, but not limited to, excreta, secreta, blood and its components, tissue and tissue fluid swabs and body parts being transported for purposes such as research, diagnosis, investigational activities, disease treatment and prevention.

3.6.2.1.5 Medical or clinical wastes are wastes derived from the veterinary treatment of animals, the medical treatment of humans or from bio-research.

3.6.2.2 Classification of Infectious Substances

3.6.2.2.1 Infectious substances must be classified in Division 6.2 and assigned to UN 2814, UN 2900, UN 3291, UN 3373 or UN 3549, as appropriate.

3.6.2.2.2 Infectious substances are divided into the following categories:

3.6.2.2.2.1 Category A: An infectious substance which is transported in a form that, when exposure to it occurs, is capable of causing permanent disability, life-threatening or fatal disease in otherwise healthy humans or animals. Indicative examples of substances that meet these criteria are given in Table 3.6.D.

Note:

An exposure occurs when an infectious substance is released outside of the protective packaging, resulting in physical contact with humans or animals.

- (a) Infectious substances meeting these criteria which cause disease in humans or both in humans and animals must be assigned to UN 2814. Infectious substances which cause disease only in animals must be assigned to UN 2900.
- (b) Assignment to UN 2814 or UN 2900 must be based on the known medical history and symptoms of the source human or animal, endemic local conditions, or professional judgement concerning individual circumstances of the source human or animal.

Notes:

1. *The proper shipping name for UN 2814 is **Infectious substance, affecting humans**. The proper shipping name for UN 2900 is **Infectious substance, affecting animals** only.*
2. *The following table is not exhaustive. Infectious substances, including new or emerging pathogens, which do not appear in the table but which meet the same criteria must be assigned to Category A. In addition, if there is doubt as to whether or not a substance meets the criteria it must be included in Category A.*
3. *In the following table, the micro-organisms written in italics are bacteria or fungi.*

TABLE 3.6.D
Indicative Examples of Infectious Substances Included in Category A in
Any Form Unless Otherwise Indicated (3.6.2.2.1)

UN Number and Proper Shipping Name	Micro-organism
UN 2814 Infectious substance affecting humans	<i>Bacillus anthracis</i> (cultures only) <i>Brucella abortus</i> (cultures only) <i>Brucella melitensis</i> (cultures only) <i>Brucella suis</i> (cultures only) <i>Burkholderia mallei</i> – <i>Pseudomonas mallei</i> –Glanders (cultures only) <i>Burkholderia pseudomallei</i> – <i>Pseudomonas pseudomallei</i> (cultures only) <i>Chlamydia psittaci</i> –avian strains (cultures only) <i>Clostridium botulinum</i> (cultures only) <i>Coccidioides immitis</i> (cultures only) <i>Coxiella burnetii</i> (cultures only) Crimean-Congo haemorrhagic fever virus Dengue virus (cultures only) Eastern equine encephalitis virus (cultures only) <i>Escherichia coli</i> , verotoxigenic (cultures only) Ebola virus Flexal virus <i>Francisella tularensis</i> (cultures only) Guanarito virus Hantaan virus Hantavirus causing hemorrhagic fever with renal syndrome Hendra virus Hepatitis B virus (cultures only) Herpes B virus (cultures only) Human immunodeficiency virus (cultures only) Highly pathogenic avian influenza virus (cultures only) Japanese Encephalitis virus (cultures only) Junin virus Kyasanur Forest disease virus Lassa virus Machupo virus Marburg virus Monkeypox virus <i>Mycobacterium tuberculosis</i> (cultures only) Nipah virus Omsk haemorrhagic fever virus Poliovirus (cultures only) Rabies virus (cultures only) <i>Rickettsia prowazekii</i> (cultures only) <i>Rickettsia rickettsii</i> (cultures only) Rift Valley fever virus (cultures only) Russian spring-summer encephalitis virus (cultures only)

TABLE 3.6.D
Indicative Examples of Infectious Substances Included in Category A in
Any Form Unless Otherwise Indicated (3.6.2.2.1) (continued)

UN Number and Proper Shipping Name	Micro-organism
	Sabia virus <i>Shigella dysenteriae</i> type 1 (cultures only) Tick-borne encephalitis virus (cultures only) Variola virus Venezuelan equine encephalitis virus (cultures only) West Nile virus (cultures only) Yellow fever virus (cultures only) <i>Yersinia pestis</i> (cultures only)
UN 2900 Infectious substances affecting animals	African swine fever virus (cultures only) Avian paramyxovirus Type 1–Velogenic Newcastle disease virus (cultures only) Classical swine fever virus (cultures only) Foot and mouth disease virus (cultures only) Goatpox virus (cultures only) Lumpy skin disease virus (cultures only) <i>Mycoplasma mycoides</i> –Contagious bovine pleuropneumonia (cultures only) Peste des petits ruminants virus (cultures only) Rinderpest virus (cultures only) Sheep-pox virus (cultures only) Swine vesicular disease virus (cultures only) Vesicular stomatitis virus (cultures only)

3.6.2.2.2 Category B: An infectious substance which does not meet the criteria for inclusion in Category A. Infectious substances in Category B must be assigned to UN 3373.

Note:

*The proper shipping name of UN 3373 is **Biological substance Category B**.*

3.6.2.2.3 Exceptions

3.6.2.2.3.1 Substances which do not contain infectious substances or substances which are unlikely to cause disease in humans or animals are not subject to these Regulations unless they meet the criteria for inclusion in another class.

3.6.2.2.3.2 Substances containing micro-organisms, which are non-pathogenic to humans or animals are not subject to these Regulations unless they meet the criteria for inclusion in another class.

3.6.2.2.3.3 Substances in a form that any present pathogens have been neutralized or inactivated such that they no longer pose a health risk are not subject to these Regulations unless they meet the criteria for inclusion in another class.

3.6.2.2.3.4 Environmental samples (including food and water samples), which are not considered to pose a significant risk of infection are not subject to these Regulations, unless they meet the criteria for inclusion in another class.

3.6.2.2.3.5 Dried blood spots, collected by applying a drop of blood onto absorbent material are not subject to these Regulations.

3.6.2.2.3.6 Faecal occult blood screening samples are not subject to these Regulations.

3.6.2.2.3.7 Blood or blood components which have been collected for the purposes of transfusion or for the preparation of blood products to be used for transfusion or transplantation and any tissues or organs intended for use in transplantation as well as samples drawn in connection with such purposes are not subject to these Regulations.

3.6.2.2.3.8 Patient specimens for which there is minimal likelihood that pathogens are present are not subject to other provisions of these Regulations provided:

- (a) The specimen must be packed in a packaging which will prevent any leakage and which is marked with the words "Exempt human specimen" or "Exempt animal specimen," as appropriate;
- (b) The packaging must consist of three components:
 1. a leak-proof primary receptacle(s);
 2. a leak-proof secondary packaging; and
 3. an outer packaging of adequate strength for its capacity, mass and intended use and with at least one surface having minimum dimensions of 100 mm × 100 mm;

- (c) For liquids, absorbent material in sufficient quantity to absorb the entire contents must be placed between the primary receptacle(s) and the secondary packaging so that, during transport, any release or leak of a liquid substance will not reach the outer packaging and will not compromise the integrity of the cushioning material;
- (d) When multiple fragile primary receptacles are placed in a single secondary packaging, they must be either individually wrapped or separated to prevent contact between them.
- (e) If refrigerated or frozen specimens are to be transported the following conditions must be met:
 - 1. when dry ice or liquid nitrogen is used to keep specimens cold, all applicable requirements of these Regulations must be met. When used, ice or dry ice must be placed between the secondary packaging and the outer packaging. Interior supports must be provided to secure the secondary packaging in the original position after the ice or dry ice has dissipated. If ice is used, the outside packaging must be leakproof. If carbon dioxide, solid (dry ice) is used, the packaging must be designed and constructed to permit the release of carbon dioxide gas to prevent a build-up of pressure that could rupture the packagings;
 - 2. the primary receptacle and the secondary packaging must maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost.

Note:

In determining whether a patient specimen has a minimal likelihood that pathogens are present, an element of professional judgment is required to determine if a substance is exempt under this paragraph. That judgment should be based on the known medical history, symptoms and individual circumstances of the source, human or animal and endemic local conditions. Examples of specimens which may be transported under this paragraph include the blood or urine tests to monitor cholesterol levels, blood glucose levels, hormone levels, or prostate specific antigens (PSA); tests required to monitor organ function such as heart, liver or kidney function for humans or animals with non-infectious diseases, or therapeutic drug monitoring; tests conducted for insurance or employment purposes and are intended to determine the presence of drugs or alcohol; pregnancy tests; biopsies to detect cancer; and antibody detection in humans or animals in the absence of any concern for infection (e.g. evaluation of vaccine induced immunity, diagnosis of autoimmune disease, etc.).

3.6.2.2.3.9 Medical devices or equipment potentially contaminated with or containing infectious substances which are being transported for disinfection, cleaning, sterilization, repair, or equipment evaluation are not subject to the provisions of these Regulations when carried as cargo if packed in packagings designed and constructed in such a way that, under normal conditions of transport, they cannot break, be punctured or leak their

contents. Packagings must be designed to meet the construction requirements listed in 6.2.

3.6.2.2.3.9.1 Medical devices or equipment must be drained of free liquid to the extent practicable. They must be packed in a strong rigid outer packaging fitted with sufficient cushioning material to prevent movement within the outer packaging. These packagings must meet the general packing requirements of 5.0.2.4.1, 5.0.2.6.1.1, 5.0.2.6.1.2 and 5.0.2.7.1. If the outer packaging is not liquid tight and the medical devices or equipment are contaminated with or contain liquid infectious substances, a means of containing the liquid in the event of leakage must be provided in the form of a leakproof liner, plastic bag or other equally effective means of containment. These packages must be capable of retaining the medical devices and equipment when dropped from a height of 1.2 m.

Note:

A packaging's capability of retaining medical devices or equipment when dropped from a height of 1.2 m should be determined through testing a sample package as prepared for transport or through alternative means such as non-destructive testing and engineering analysis, testing with an article of similar mass and size, or other equivalent means.

3.6.2.2.3.9.2 Packages must be marked "Used Medical Device" or "Used Medical Equipment". When packages are placed in an overpack this mark must be reproduced on the outside of the overpack unless the mark remains visible.

3.6.2.2.3.9.3 This exception does not apply to:

- (a) medical waste (UN 3291 and UN 3549);
- (b) medical devices or equipment contaminated with or containing infectious substances in Category A (UN 2814 or UN 2900); and
- (c) medical devices or equipment contaminated with or containing other dangerous goods that meet the definition of another hazard class.

3.6.2.3 Biological Products

3.6.2.3.1 For the purposes of these Regulations, biological products are divided into the following groups:

- (a) those which are manufactured and packaged in accordance with the requirements of appropriate national authorities and transported for the purposes of final packaging or distribution and use for personal health care by medical professionals or individuals. Substances in this group are not subject to these Regulations;
- (b) those which do not fall under paragraph (a) and are known or reasonably believed to contain infectious substances and which meet the criteria for inclusion in Category A or Category B. Substances in this group must be assigned to UN 2814, UN 2900 or UN 3373, as appropriate.

Note:

Some licensed biological products may present a biohazard only in certain parts of the world. In that case, competent authorities may require these biological products to be in compliance with local

requirements for infectious substances or may impose other restrictions.

3.6.2.4 Genetically Modified Micro-organisms and Organisms

Genetically modified micro-organisms not meeting the definition of an infectious substance must be classified according to Subsection 3.9.

3.6.2.5 Medical or Clinical Wastes

3.6.2.5.1 Medical or clinical wastes containing:

- (a) Category A infectious substances must be assigned to UN 2814, UN 2900 or UN 3549, as appropriate. Solid medical waste containing Category A infectious substances generated from the medical treatment of humans or veterinary treatment of animals may be assigned to UN 3549. The UN 3549 entry must not be used for waste from bioresearch or liquid waste;
- (b) Category B infectious substances must be assigned to UN 3291.

3.6.2.5.2 Medical or clinical wastes which are reasonably believed to have a low probability of containing infectious substances must be assigned to UN 3291. For the assignment, international, regional or national waste catalogues may be taken into account.

Note:

*The proper shipping name for UN 3291 is **Biomedical waste, n.o.s., Clinical waste, unspecified, n.o.s. or Medical waste, n.o.s. or Regulated medical waste, n.o.s.***

3.6.2.5.3 Decontaminated medical or clinical wastes which previously contained infectious substances are not subject to these Regulations unless they meet the criteria for inclusion in another class.

3.6.2.6 Infected Animals

3.6.2.6.1 Infected Live Animals

Live animals must not be used to consign infectious substances unless such a substance cannot be consigned by any other means. A live animal that has been intentionally infected and is known or suspected to contain an infectious substance may only be transported by air under terms and conditions approved by the appropriate national authority of the States of origin, transit, destination and operator in accordance with the Supplement to the Technical Instructions (Part S-1;2).

3.6.2.7 Patient Specimens

Patient specimens must be assigned to UN 2814, UN 2900 or UN 3373 as appropriate except if they comply with 3.6.2.2.3.

3.7 Class 7—Radioactive Material

STATE VARIATIONS: RUG-01, USG-10

3.7.1 Definition

Radioactive material means any material containing radionuclides where both the activity concentration and the total activity in the consignment exceed the values specified in 10.3.2.

These Regulations do not apply to any of the following:

- (a) radioactive material implanted or incorporated into a person or live animal for diagnosis or treatment;
- (b) radioactive material in or on a person who is to be transported for medical treatment because the person has been subject to accidental or deliberate intake of radioactive material or to contamination, taking into account the necessary radiological protection measures with respect to other passengers and crew, subject to approval by the operator;

Note:

Guidance material may be found on www.icao.int/safety/DangerousGoods/Pages/Guidance-Material.aspx.

- (c) radioactive material in consumer products which have received regulatory approval, following their sale to the end user;
- (d) natural material and ores containing naturally occurring radionuclides (which may have been processed), provided the activity concentration of the material does not exceed 10 times the values specified in Table 10.3.A or calculated in accordance with 10.3.2.2(a) to 10.3.2.5. For natural materials and ores containing naturally occurring radionuclides that are not in secular equilibrium, the calculation of the activity concentration must be performed in accordance with 10.3.2.4;
- (e) non-radioactive solid objects with radioactive substances present on any surfaces in quantities not in excess of the limit specified in the definition of contamination in Appendix A.

3.8 Class 8—Corrosives

3.8.1 Definition

3.8.1.1 Corrosive substances are substances which, by chemical action, will cause irreversible damage to the skin or, in the case of leakage will materially damage or even destroy, other goods or the means of transport.

3.8.1.2 For substances and mixtures that are corrosive to skin, general classification provisions are provided in 3.8.2. Skin corrosion refers to the production of irreversible damage to the skin, namely, visible necrosis through the epidermis and into the dermis occurring after exposure to a substance or mixture.

3.8.1.3 Liquids and solids which may become liquid during transport, which are judged not to be skin corrosive must still be considered for their potential to cause corrosion to certain metal surfaces in accordance with the criteria in 3.8.3.3.3(b).

3.8.2 General Classification Provisions

3.8.2.1 Assignment of substances in Class 8 listed in Table 4.2 to the packing groups referred to in 3.0.3 has been made on the basis of experience, taking into account such additional factors as inhalation risk (see 3.8.2.3) and reactivity with water, including the formation of hazardous decomposition products.

3.8.2.2 New substances and mixtures, can be assigned to packing groups on the basis of the length of time of contact necessary to produce irreversible damage of intact skin tissue in accordance with the criteria in 3.8.3. Alternatively, for mixtures, the criteria in 3.8.4 can be used.

3.8.2.3 Substances or mixtures meeting the criteria of Class 8 having an inhalation toxicity of dusts and mists (LC₅₀) in the range of Packing Group I but toxicity through oral ingestion or dermal contact only in the range of Packing Group III or less must be allocated to Class 8 (see Note under 3.6.1.4).

3.8.3 Packing Group Assignment for Substances and Mixtures

3.8.3.1 Existing human and animal data including information from single or repeated exposure must be the first line of evaluation, as they give information directly relevant to effects on the skin.

3.8.3.2 In assigning the packing group in accordance with 3.8.2.2:

3.8.3.2.1 Account must be taken of human experience in instances of accidental exposure. In the absence of human experience, the classification must be based on the data obtained from experiments in accordance with *Organization for Economic Co-operation and Development (OECD) Guidelines for Testing of Chemicals No. 404, Acute Dermal Irritation/Corrosion, 2015, No. 435, In Vitro Membrane Barrier Test Method for Skin Corrosion, 2015, No. 431, In Vitro Skin Corrosion: Reconstructed Human Epidermis (RHE) Test Method, 2016 or No. 430, In Vitro Skin Corrosion: Transcutaneous Electrical Resistance Test (TER) Test Method, 2015.*

△ **3.8.3.2.2** A substance or mixture which is determined not to be corrosive in accordance with OECD Guideline for the Testing of Chemicals No. 404, No. 435, No. 431 or No. 430 or non-classified in accordance with No. 439, In Vitro Skin Irritation: Reconstructed Human Epidermis Test Method, 2015, may be considered not to be corrosive to skin for the purposes of these Regulations without further testing.

△ **3.8.3.2.3** If the test results indicate that the substance or mixture is corrosive and not assigned to Packing Group I, but the test method does not allow discrimination between Packing Groups II and III, it must be assigned to Packing Group II. If the test results indicate that the substance or mixture is corrosive, but the test method does not allow discrimination between packing groups, it must be

assigned to Packing Group I if no other test results indicate a different packing group.

3.8.3.3 Packing groups are assigned to corrosive substances in accordance with the criteria in 3.8.3.3.1 through 3.8.3.3.3.

3.8.3.3.1 Packing Group I. Substances that cause irreversible damage of intact skin tissue within an observation period of up to 60 minutes starting after the exposure time of 3 minutes or less.

3.8.3.3.2 Packing Group II. Substances that cause irreversible damage of intact skin tissue within an observation period of up to 14 days starting after the exposure time of more than 3 minutes but not more than 60 minutes.

3.8.3.3.3 Packing Group III.

Substances that:

(a) cause irreversible damage of intact skin tissue within an observation period of up to 14 days starting after the exposure time of more than 60 minutes but not more than 4 hours; or

△ (b) are judged not to cause irreversible damage of intact skin tissue but which exhibit a corrosion rate either on steel or aluminium surfaces exceeding 6.25 mm a year at a test temperature of 55°C when tested on both materials. For the purposes of testing steel, type S235JR+CR (1.0037 resp. St 37-2), S275J2G3+CR (1.0144 resp. St 44-3), ISO 3574, Unified Numbering System (UNS) G10200 or SAE 1020 must be used; for the purposes of testing aluminium, non-clad types 7075-T6 or AZ5GU-T6, must be used. An acceptable test is prescribed in the UN Manual of Tests and Criteria, Part III, Section 37.

Note:

Where an initial test on either steel or aluminium indicates the substance being tested is corrosive the follow up test on the other metal is not required.

TABLE 3.8.A
Class 8—Packing Group Assignment
based on Corrosivity (3.8.3)

Packing Group	Exposure Time	Observation Time	Effect
I	≤ 3 min	≤ 60 min	Irreversible damage of intact skin
II	> 3 min ≤ 60 min	≤ 14 d	Irreversible damage of intact skin
III	> 60 min ≤ 4 h	≤ 14 d	Irreversible damage of intact skin
III	—	—	Corrosion rate on steel/aluminium > 6.25 mm a year at a test temperature of 55°C

Note:

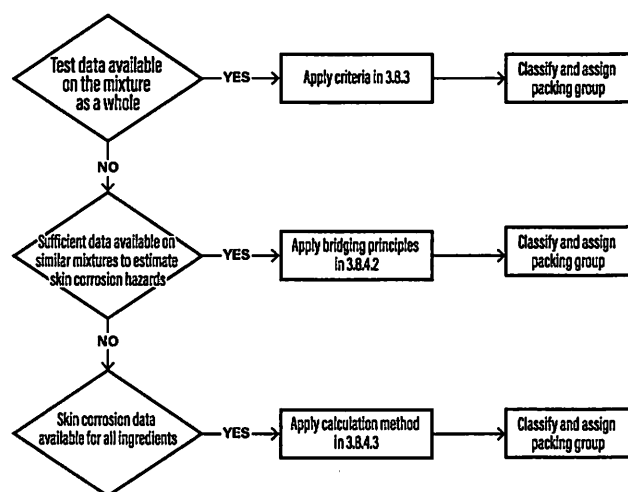
h = hours, d = days.

3.8.4 Alternative Packing Group Assignment Methods for Mixtures: Step-wise Approach

3.8.4.1 General Provisions

For mixtures it is necessary to obtain or derive information that allows the criteria to be applied to the mixture for the purpose of classification and assignment of packing groups. The approach to classification and assignment of packing groups is tiered, and is dependent upon the amount of information available for the mixture itself, for similar mixtures and/or for its ingredients. The flow chart of Figure 3.8.B below outlines the process to be followed:

FIGURE 3.8.B
Step-Wise Approach to Classify and Assign a Packing Group to a Corrosive Mixture (3.8.4.1)



3.8.4.2 Bridging Principles

3.8.4.2.1 Where a mixture has not been tested to determine its skin corrosion potential, but there are sufficient data on both the individual ingredients and similar tested mixtures to adequately classify and assign a packing group for the mixture, these data will be used in accordance with the following bridging principles. This ensures that the classification process uses the available data to the greatest extent possible in characterizing the hazards of the mixture.

- (a) **Dilution:** If a tested mixture is diluted with a diluent which does not meet the criteria for Class 8 and does not affect the packing group of other ingredients, then the new diluted mixture may be assigned to the same packing group as the original tested mixture.

Note:

In certain cases, diluting a mixture or substance may lead to an increase in the corrosive properties. If this is the case, this bridging principle cannot be used.

- (b) **Batching:** The skin corrosion potential of a tested production batch of a mixture can be assumed to be substantially equivalent to that of another untested production batch of the same commercial product

when produced by or under the control of the same manufacturer, unless there is reason to believe there is significant variation such that the skin corrosion potential of the untested batch has changed. If the latter occurs, a new classification is necessary.

- (c) **Concentration of mixtures of Packing Group I:** If a tested mixture meeting the criteria for inclusion in Packing Group I is concentrated, the more concentrated untested mixture may be assigned to Packing Group I without additional testing.
- (d) **Interpolation within one packing group:** For three mixtures (A, B and C) with identical ingredients, where mixtures A and B have been tested and are in the same skin corrosion packing group, and where untested mixture C has the same Class 8 ingredients as mixtures A and B but has concentrations of Class 8 ingredients intermediate to the concentrations in mixtures A and B, then mixture C is assumed to be in the same skin corrosion packing group as A and B.
- (e) **Substantially similar mixtures:** Given the following:
1. two mixtures: (A+B) and (C+B);
 2. the concentration of ingredient B is the same in both mixtures;
 3. the concentration of ingredient A in mixture (A+B) equals the concentration of ingredient C in mixture (C+B);
 4. data on skin corrosion for ingredients A and C are available and substantially equivalent, i.e. they are the same skin corrosion packing group and do not affect the skin corrosion potential of B.

If mixture (A+B) or (C+B) is already classified based on test data, then the other mixture may be assigned to the same packing group.

3.8.4.3 Calculation Method Based on the Classification of the Substances

3.8.4.3.1 Where a mixture has not been tested to determine its skin corrosion potential, nor is sufficient data available on similar mixtures, the corrosive properties of the substances in the mixture must be considered to classify and assign a packing group.

Applying the calculation method is only allowed if there are no synergistic effects that make the mixture more corrosive than the sum of its substances. This restriction applies only if Packing Group II or III would be assigned to the mixture.

3.8.4.3.2 When using the calculation method, all Class 8 ingredients present at a concentration of $\geq 1\%$ must be taken into account, or $< 1\%$ if these ingredients are still relevant for classifying the mixture to be corrosive to skin.

3.8.4.3.3 To determine whether a mixture containing corrosive substances must be considered a corrosive mixture and to assign a packing group, the calculation method in the flow chart in Figure 3.8.C must be applied.

3.8.4.3.4 When a specific concentration limit (SCL) is assigned to a substance following its entry in Table 4.2 or in a special provision, this limit must be used instead of the generic concentration limits (GCL). This appears

where 1% is used in the first step for the assessment of the Packing Group I substances, and where 5% is used for the other steps respectively in Figure 3.8.C.

3.8.4.3.5 For this purpose, the summation formula for each step of the calculation method must be adapted. This means that, where applicable, the generic concentration limit must be substituted by the specific concentration limit assigned to the substance(s) (SCL_i), and the adapted formula is a weighted average of the different concentration limits assigned to the different substances in the mixture:

$$\frac{PGx_1}{GCL} + \frac{PGx_2}{SCL_2} + \dots + \frac{PGx_i}{SCL_i} \geq 1$$

Where:

PG x_i = concentration of substance 1, 2 ...i in the mixture, assigned to packing group x (I, II or III)

GCL = generic concentration limit

SCL_i = specific concentration limit assigned to substance i

The criterion for a packing group is fulfilled when the result of the calculation is ≥ 1 . The generic concentration limits to be used for the evaluation in each step of the calculation method are those found in Figure 3.8.C.

Examples for the application of the above formula can be found in the note below.

Note:

Examples for the application of the above formula:

Example 1: A mixture contains one corrosive substance in a concentration of 5% assigned to Packing Group I without a specific concentration limit:

Calculation for Packing Group I:

$$\frac{5}{5 (GCL)} = 1$$

assign to Class 8, Packing Group I;

Example 2: A mixture contains three substances corrosive to skin; two of them (A and B) have specific concentration limits; for the third one (C) the generic concentration limits applies. The rest of the mixture needs not to be taken into consideration:

Substance X in the mixture and its packing group assignment within class 8	Concentration (conc) in the mixture	Specific concentration limit (SCL) for Packing Group I	Specific concentration limit (SCL) for Packing Group II	Specific concentration limit (SCL) for Packing Group III
A, assigned to Packing Group I	3%	30%	none	none
B, assigned to Packing Group I	2%	20%	10%	none
C, assigned to Packing Group III	10%	none	none	none

Calculation for Packing Group I:

$$\frac{3 (conc A)}{30 (SCL PG I)} + \frac{2 (conc B)}{20 (SCL PG I)} = 0.2 < 1$$

The criterion for Packing Group I is not fulfilled.

Calculation for Packing Group II:

$$\frac{3 (conc A)}{5 (GCL PG II)} + \frac{2 (conc B)}{10 (SCL PG II)} = 0.8 < 1$$

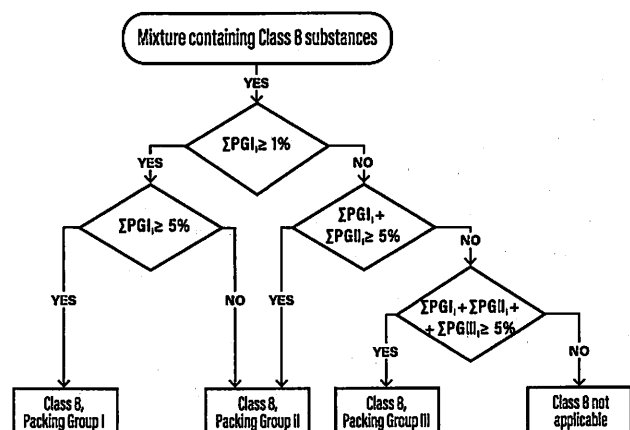
The criterion for Packing Group II is not fulfilled.

Calculation for Packing Group III:

$$\frac{3 (conc A)}{5 (GCL PG III)} + \frac{2 (conc B)}{5 (GCL PG III)} + \frac{10 (conc C)}{5 (GCL PG III)} = 3 \geq 1$$

The criterion for Packing Group III is fulfilled; the mixture must be assigned to Class 8, PG III.

FIGURE 3.8.C
Calculation Method to Assign a Packing Group to a Corrosive Mixture (3.8.4.3)



3.8.5 Substances Forbidden for Transport

Chemically unstable substances of Class 8 are forbidden for transport unless the necessary precautions have been taken to prevent the possibility of a dangerous decomposition or polymerization under normal conditions of transport. For the precautions necessary to prevent polymerization, see Special Provision A209. To this end particular care must be taken to ensure that receptacles and tanks do not contain any substances liable to promote these reactions.

3.9 Class 9—Miscellaneous Dangerous Substances and Articles, Including Environmentally Hazardous Substances

STATE VARIATIONS: DEG-05, NLG-04, USG-11/16

3.9.1 Definition

3.9.1.1 Class 9 Substances and Articles

Articles and substances which, during air transport, present a danger not covered by other classes.

3.9.2 Assignment to Class 9

Class 9 includes the following articles and substances:

3.9.2.1 Aviation Regulated Solid or Liquid

3.9.2.1.0 Assigned entries:

- UN 3334 Aviation regulated liquid, n.o.s.
- UN 3335 Aviation regulated solid, n.o.s.

3.9.2.1.1 Any material, which has narcotic, noxious, irritating or other properties such that, in the event of spillage or leakage on an aircraft, could cause extreme annoyance or discomfort to crew members so as to prevent the correct performance of assigned duties. The materials included under this proper shipping name must not meet any of the definitions for Classes 1 through 8.

3.9.2.2 Magnetized Material

3.9.2.2.0 Assigned entry:

- UN 2807 Magnetized material

3.9.2.2.1 Any material which, when packed for air transport, has a maximum magnetic field strength sufficient to cause a compass deflection of more than 2 degrees at a distance of 2.1 m from any point on the surface of the assembled package. The magnetic field strength at the compass producing a 2 degree deflection is taken to be 0.418 A/m (0.00525 Gauss).

3.9.2.2.2 The magnetic field strength must be measured with a magnetic compass sensitive enough to read a 2

degree variation, preferably in 1 degree increments or finer, or using a Gauss meter having a sensitivity sufficient to measure magnetic fields greater than 0.0005 Gauss within a tolerance of ± 5 per cent, or by an equivalent means.

3.9.2.2.3 Compass measurements must be taken in an area free from magnetic interference other than the earth's magnetic field. When using a compass, the material and the compass must be aligned in an East/West direction. Gauss meter measurements must be in accordance with the manufacturer's instructions. Measurements are taken while the packaged material is rotated through 360 degrees in its horizontal plane while maintaining a constant distance (2.1 m or 4.6 m as referred to in Packing Instruction 953) between the measuring device and any point on the outside surface of the package. Shielding may be used to reduce the package's magnetic strength.

Note:

Masses of ferro-magnetic metals such as automobiles, automobile parts, metal fencing, piping and metal construction material, even if not meeting the definition of magnetized materials may affect aircraft compasses, as may packages or items which individually do not meet the definition of magnetized materials but cumulatively may have a magnetic field strength of a magnetized material.

3.9.2.3 Elevated Temperature Substances

3.9.2.3.0 Assigned entries:

- UN 3257 Elevated temperature liquid, n.o.s.
- UN 3258 Elevated temperature solid, n.o.s.

3.9.2.3.1 Substances that are transported or offered for transport at temperatures equal to or exceeding 100°C in a liquid state or at temperatures equal or exceeding 240°C in a solid state (these substances may only be carried under the provisions of 2.1.2).

3.9.2.4 Environmentally Hazardous Substances

3.9.2.4.0 Substances or mixtures dangerous to the aquatic environment not presenting a danger covered by other classes, must be assigned to packing group III and designated:

- UN 3077 Environmentally hazardous substance, solid, n.o.s.; or
- UN 3082 Environmentally hazardous substance, liquid, n.o.s.

3.9.2.4.1 Environmentally hazardous substances (aquatic environment) are those that meet the criteria in 2.9.3 of the UN Model Regulations or that meet criteria in national or international regulations established by the appropriate national authority in the State of origin, transit or destination of the consignment. The detailed classification categories and criteria for environmentally hazardous substances (aquatic environment) as set out in 2.9.3 of the UN Model Regulations can be found at <https://www.iata.org/dgr-updates>

3.9.2.5 Genetically Modified Micro-Organisms (GMMOs) or Genetically Modified Organisms (GMOs)

3.9.2.5.0 Assigned entries:

- UN 3245 Genetically modified micro-organisms or Genetically modified organisms

3.9.2.5.1 Genetically modified micro-organisms (GMMOs) and genetically modified organisms (GMOs) are micro-organisms and organisms in which genetic material has been purposely altered through genetic engineering in a way that does not occur naturally.

3.9.2.5.2 Genetically modified organisms and micro-organisms which do not meet the definition of toxic or infectious substances must be assigned to UN 3245.

3.9.2.5.3 GMMOs or GMOs are not subject to these Regulations when authorised for use by the appropriate national authorities of the States of origin, transit and destination.

3.9.2.5.4 Genetically modified live animals must be transported under terms and conditions of the appropriate national authorities of the States of origin and destination.

3.9.2.5.5 COVID-19 vaccines containing GMOs or GMMOs, including those in clinical trials, are not subject to these Regulations.

3.9.2.6 Lithium Batteries

3.9.2.6.0 Assigned entries:

- UN 3090 Lithium metal batteries
- UN 3091 Lithium metal batteries contained in equipment or Lithium metal batteries packed with equipment
- UN 3480 Lithium ion batteries
- UN 3481 Lithium ion batteries contained in equipment or Lithium ion batteries packed with equipment
- UN 3536 Lithium batteries installed in cargo transport unit

3.9.2.6.1 Cells and batteries, cells and batteries contained in equipment, or cells and batteries packed with equipment, containing lithium in any form must be assigned to UN 3090, UN 3091, UN 3480 or UN 3481, as appropriate. They may be transported under these entries if they meet the following provisions:

- each cell or battery is of the type proved to meet the requirements of each test of the UN *Manual of Tests and Criteria*, Part III, subsection 38.3. Cells and batteries manufactured according to a type meeting the requirements of subsection 38.3 of the UN *Manual of Tests and Criteria*, Revision 3, Amendment 1 or any subsequent revision and amendment applicable at the date of the type testing may continue to be transported, unless otherwise provided in these Regulations. Cell and battery types only meeting the requirements of the UN *Manual of*

Tests and Criteria, Revision 3, are no longer valid. However, cells and batteries manufactured in conformity with such types before 1 July 2003 may continue to be transported if all other applicable requirements are fulfilled.

Note:

Batteries, including those which have been refurbished or otherwise altered, must be of a type proved to meet the testing requirements of the Manual of Tests and Criteria, Part III, subsection 38.3, irrespective of whether the cells of which they are composed are of a tested type.

- each cell and battery incorporates a safety venting device or is designed to preclude a violent rupture under conditions normally incident to transport;
- each cell and battery is equipped with an effective means of preventing external short circuits;
- each battery containing cells or series of cells connected in parallel is equipped with effective means as necessary to prevent dangerous reverse current flow (e.g., diodes, fuses, etc.);
- cells and batteries must be manufactured under a quality management program that includes:
 - a description of the organizational structure and responsibilities of personnel with regard to design and product quality;
 - the relevant inspection and test, quality control, quality assurance and process operation instructions that will be used;
 - process controls that should include relevant activities to prevent and detect internal short circuit failure during manufacture of cells;
 - quality records, such as inspection reports, test data, calibration data and certificates. Test data must be kept and made available to the appropriate national authority upon request;
 - management reviews to ensure the effective operation of the quality management program;
 - a process for control of documents and their revision;
 - a means for control of cells or batteries that are not conforming to the type tested as mentioned in (a) above;
 - training programs and qualification procedures for relevant personnel; and
 - procedures to ensure that there is no damage to the final product.

Note:

In house quality management programs may be accepted. Third party certification is not required, but the procedures listed in 1. to 9. above must be properly recorded and traceable. A copy of the quality management program must be made available to the appropriate national authority upon request.

(f) lithium batteries, containing both primary lithium metal cells and rechargeable lithium ion cells, that are not designed to be externally charged (see Special Provision A213) must meet the following conditions:

1. the rechargeable lithium ion cells can only be charged from the primary lithium metal cells;
2. overcharge of the rechargeable lithium ion cells is precluded by design;
3. the battery has been tested as a lithium primary battery;
4. component cells of the battery must be of a type proved to meet the respective testing requirements of the UN Manual of Tests and Criteria, Part III, sub-section 38.3.

△ (g) except for button cells installed in equipment (including circuit boards), manufacturers and subsequent distributors of cells or batteries manufactured after 30 June 2003 must make available the test summary as specified in the UN Manual of Tests and Criteria, Part III, sub-section 38.3, paragraph 38.3.5.

3.9.2.7 Substances which, on Inhalation as Fine Dust, may Endanger Health

3.9.2.7.0 Assigned entries:

- UN 2212 **Asbestos, amphibole** (amosite, tremolite, actinolite, anthophyllite, crocidolite)
- UN 2590 **Asbestos, chrysotile**

3.9.2.8 Capacitors

3.9.2.8.0 Assigned entries:

- UN 3499 **Capacitor, electric double layer**
- UN 3508 **Capacitor asymmetric**

3.9.2.9 Substances Evolving Flammable Vapour

3.9.2.9.0 Assigned entries:

- UN 2211 **Polymeric beads, expandable**
- UN 3314 **Plastics moulding compound**

3.9.2.10 Life-Saving Appliances

3.9.2.10.0 Assigned entries:

- UN 2990 **Life-saving appliances, self-inflating**
- UN 3072 **Life-saving appliances, not self-inflating**
- UN 3268 **Safety devices**

3.9.2.11 Substances and Articles which, in the Event of Fire, may Form Dioxins

3.9.2.11.0 Assigned entries:

- UN 2315 **Polychlorinated biphenyls, liquid**
- UN 3151 **Polyhalogenated biphenyls, liquid or Halogenated monomethyldiphenylmethanes, liquid or Polyhalogenated terphenyls, liquid**
- UN 3152 **Polyhalogenated biphenyls, solid, Polyhalogenated terphenyls, solid or Halogenated monomethyldiphenylmethanes, solid**
- UN 3432 **Polychlorinated biphenyls, solid**

3.9.2.12 Other Substances or Articles Presenting a Danger During Transport, but not Meeting the Definitions of Another Class

3.9.2.12.0 Assigned entries:

- UN 1841 **Acetaldehyde ammonia**
- UN 1845 **Carbon dioxide, solid or Dry ice**
- UN 1931 **Zinc dithionite or Zinc hydrosulphite**
- UN 1941 **Dibromodifluoromethane**
- UN 1990 **Benzaldehyde**
- UN 2071 **Ammonium nitrate based fertilizer**
- UN 2216 **Fish meal, stabilized or Fish scrap, stabilized**
- UN 2969 **Castor beans or Castor flake or Castor meal or Castor pomace**
- UN 3166 **Vehicle, flammable gas powered or Vehicle, flammable liquid powered or Vehicle, fuel cell, flammable gas powered or Vehicle, fuel cell, flammable liquid powered**
- UN 3171 **Battery-powered equipment or Battery-powered vehicle**
- UN 3316 **Chemical kit or First aid kit**
- UN 3359 **Fumigated cargo transport unit**
- UN 3363 **Dangerous goods in articles or Dangerous goods in apparatus or Dangerous goods in machinery**
- UN 3509 **Packagings, discarded, empty, uncleaned**
- UN 3530 **Engines, internal combustion or Machinery, internal combustion**
- ID 8000 **Consumer commodity (see A112)**

3.10 Classification of Substances with Multiple Hazards

When a substance is not specifically listed by name in Subsection 4.2—List of Dangerous Goods and has more than one hazard, the primary and subsidiary hazards must be determined in accordance with the criteria presented in this subsection.

3.10.1 Precedence of Hazards Table

3.10.1.1 Application

When the two hazards fall within Classes 3, 4 or 8 or Divisions 5.1 or 6.1, or to assign the appropriate entry for articles containing dangerous goods, n.o.s., UN 3537 to UN 3548, see 3.12, Table 3.10.A must be used to determine which of the two hazards must be regarded as the primary hazard. The class or division, which appears at the intersection of the relevant line and column, is the primary hazard and the other class or division is the subsidiary hazard. The correct packing group to be used is also shown at the intersection of the relevant line and column.

3.10.1.2 Packing Group

The most stringent packing group based on the different hazards of a material must then be the packing group for the substance and this is shown next to the primary hazard at the intersection of the relevant line and column.

3.10.1.3 Proper Shipping Name

The proper shipping name of a substance, when determined in accordance with Table 3.10.A must be the most appropriate n.o.s. entry in Table 4.1.A for the class or division shown as the primary hazard (see 4.1.3.2).

3.10.2 Exceptions

Substances, which meet among other hazards, the criteria for the following classes, divisions or particular types of hazards are not dealt with by Table 3.10.A since these classes, divisions or particular types of hazard always take precedence:

- (a) Classes 1, 2 and 7 (except see 3.10.3);
- (b) Divisions 5.2 and 6.2 (see also 3.10.5);
- (c) self-reactive and related substances and solid desensitized explosives of Division 4.1;
- (d) pyrophoric substances of Division 4.2;
- (e) substances of Division 6.1 with a Packing Group I inhalation toxicity. Except for substances or preparations meeting the criteria of Class 8 having an inhalation toxicity of dusts and mists (LC_{50}) in the range of Packing Group I, but toxicity through oral ingestion or dermal contact only in the range of Packing Group III or less, which must be allocated to Class 8; and
- (f) liquid desensitized explosives of Class 3.

3.10.3 Radioactive Materials

Radioactive materials having other hazardous properties must always be classified in Class 7 and the subsidiary hazard must also be identified, other than for excepted packages of radioactive materials where the other hazardous properties take precedence. For radioactive material in excepted packages, except for UN 3507, Uranium hexafluoride, radioactive material, excepted package, Special Provision A130 applies. It is also necessary to take into account the possible formation of products having additional dangerous properties by interaction of the materials with the atmosphere or with water.

3.10.4 Magnetized Material

An article, which also meets the criterion for a magnetized material, must be identified in accordance with the provisions of this subsection and in addition, as a magnetized material.

3.10.5 Infectious Substances

Infectious substances having other hazardous properties must always be classified in Division 6.2 and the greatest of the additional hazards must also be identified.

TABLE 3.10.A
Precedence of Hazards and Packing Groups for Classes 3, 4 and 8 and for
Divisions 5.1 and 6.1 (3.10.1)

Class or Division	Packing Group	4.2 II	4.2 III	4.3 I	4.3 II	4.3 III	5.1 I	5.1 II	5.1 III	6.1 (d) I	6.1 (o) I	6.1 II	6.1 III	8 (f) I	8 (s) I	8 (f) II	8 (s) II	8 (f) III	8 (s) III
3	I*			4.3, I	4.3, I	4.3, I	—	—	—	3, I	3, I	3, I	3, I	3, I	—	3, I	—	3, I	—
3	II*			4.3, I	4.3, II	4.3, II	—	—	—	3, I	3, I	3, II	3, II	8, I	—	3, II	—	3, II	—
3	III*			4.3, I	4.3, II	4.3, III	—	—	—	6.1, I	6.1, I	6.1, II	3, III**	8, I	—	8, II	—	3, III	—
4.1	II*	4.2, II	4.2, II	4.3, I	4.3, II	4.3, II	5.1, I	4.1, II	4.1, II	6.1, I	6.1, I	4.1, II	4.1, II	—	8, I	—	4.1, II	—	4.1, II
4.1	III*	4.2, II	4.2, III	4.3, I	4.3, II	4.3, III	5.1, I	4.1, II	4.1, III	6.1, I	6.1, I	6.1, II	4.1, III	—	8, I	—	8, II	—	4.1, III
4.2	II			4.3, I	4.3, II	4.3, II	5.1, I	4.2, II	4.2, II	6.1, I	6.1, I	4.2, II	4.2, II	8, I	8, I	4.2, II	4.2, II	4.2, II	4.2, II
4.2	III			4.3, I	4.3, II	4.3, III	5.1, I	5.1, II	4.2, III	6.1, I	6.1, I	6.1, II	4.2, III	8, I	8, I	8, II	8, II	4.2, III	4.2, III
4.3	I						5.1, I	4.3, I	4.3, I	6.1, I	4.3, I	4.3, I	4.3, I	4.3, I	4.3, I	4.3, I	4.3, I	4.3, I	4.3, I
4.3	II						5.1, I	4.3, II	4.3, II	6.1, I	4.3, I	4.3, II	4.3, II	8, I	8, I	4.3, II	4.3, II	4.3, II	4.3, II
4.3	III						5.1, I	5.1, II	4.3, III	6.1, I	6.1, I	6.1, II	4.3, III	8, I	8, I	8, II	8, II	4.3, III	4.3, III
5.1	I									5.1, I	5.1, I	5.1, I	5.1, I	5.1, I	5.1, I	5.1, I	5.1, I	5.1, I	5.1, I
5.1	II									6.1, I	5.1, I	5.1, II	5.1, II	8, I	8, I	5.1, II	5.1, II	5.1, II	5.1, II
5.1	III									6.1, I	6.1, I	6.1, II	5.1, III	8, I	8, I	8, II	8, II	5.1, III	5.1, III
6.1 (d)	I													8, I	6.1, I	6.1, I	6.1, I	6.1, I	6.1, I
6.1 (o)	I													8, I	6.1, I	6.1, I	6.1, I	6.1, I	6.1, I
6.1 (f)	II													8, I	6.1, I	6.1, II	6.1, II	6.1, II	6.1, II
6.1 (d)	II													8, I	6.1, I	8, II	6.1, II	6.1, II	6.1, II
6.1 (o)	II													8, I	8, I	8, II	6.1, II	6.1, II	6.1, II
6.1	III													8, I	8, I	8, II	8, II	8, III	8, III

I = liquid

s = solid

i = inhalation

d = dermal

o = oral

— = an impossible combination.

* Substances of Division 4.1 other than self-reactive substances and solid desensitized explosives and substances of Class 3 other than liquid desensitized explosives.

** For pesticides only, the primary hazard must be Division 6.1.

Note:

This table is based on the UN Precedence of Hazards Table.

3.11 Transport of Samples for Further Testing

3.11.1 Classification

When the hazard class of a substance is uncertain and it is being transported for further testing, a tentative hazard class, proper shipping name and UN number must be assigned on the basis of the shipper's knowledge of the substance and application of:

- (a) the classification criteria of these Regulations; and
- (b) the precedence of hazards given in Subsection 3.10.

The most severe packing group possible for the proper shipping name chosen must be used.

3.11.2 Proper Shipping Name

Where this provision is used, the proper shipping name must be supplemented with the word "sample" (e.g. **Flammable liquid, n.o.s., sample**). In certain instances, where a specific proper shipping name is provided for a sample of a substance considered to meet certain classification criteria (e.g. **Gas sample, non-pressurized, flammable, UN 3167**) that proper shipping name must be used. When a n.o.s. entry is used to transport the sample, the proper shipping name need not be supplemented with the technical name.

3.11.3 Limitations

Samples of the substance must be transported in accordance with the requirements applicable to the tentative assigned proper shipping name provided:

- (a) the substance is not considered to be a substance forbidden for transport by 2.1.1;
- (b) the substance is not considered to meet the criteria for Class 1 or considered to be an infectious substance or a radioactive material;
- (c) the substance is in compliance with 3.4.1.2.5 or 3.5.2.6, if it is a self-reactive substance or an organic peroxide, respectively;
- (d) the sample is transported in a combination packaging with a net weight per package not exceeding 2.5 kg; and
- (e) the sample is not packed together with other goods.

3.11.4 Samples of Energetic Materials for Testing Purposes

3.11.4.1 Samples of organic substances carrying functional groups listed in tables A6.1 and/or A6.3 in Appendix 6 (Screening Procedures) of the UN Manual of Tests and Criteria may be transported under UN 3224, **Self-reactive solid type C** or UN 3223, **Self-reactive liquid type C**, as applicable, of Division 4.1 provided that:

- (a) the samples do not contain any:
 - known explosives;
 - substances showing explosive effects in testing;
 - compounds designed with the view of producing a practical explosive or pyrotechnic effect; or

- components consisting of synthetic precursors of intentional explosives.

- (b) for mixtures, complexes or salts of inorganic oxidizing substances of Division 5.1 with organic material(s), the concentration of the inorganic oxidizing substance is:
 - less than 15%, by mass, if assigned to Packing Group I (high hazard) or II (medium hazard); or
 - less than 30%, by mass, if assigned to Packing Group III (low hazard).
- (c) available data do not allow a more precise classification;
- (d) the sample is not packed together with other goods; and
- (e) the sample is packed in accordance with Packing Instruction 459.
- (f) the proper shipping name is supplemented with the word "sample".

3.12 Classification of Articles Containing Dangerous Goods, n.o.s.

3.12.0 Articles which do not have an existing proper shipping name, and which contain only dangerous goods as a residue or as an integral element of the machinery or apparatus must be classified as follows:

- (a) where the dangerous goods meet the provisions of Packing Instruction 962: UN 3363—**Dangerous goods in apparatus** or **Dangerous goods in machinery**; or
- (b) where the net quantity of dangerous goods in the machinery or apparatus exceeds the limits of Packing Instruction 962 but contains dangerous goods permitted as limited quantities within the quantity limits specified in column 7(a) of the UN Model Regulations, see Special Provision A107; or
- (c) in accordance with paragraphs 3.12.1 to 3.12.6 of this subsection, as applicable.

3.12.1 Articles containing dangerous goods may be classified as otherwise provided by these Regulations under the proper shipping name for the dangerous goods they contain or in accordance with this section. For the purposes of this section "article" means machinery, apparatus or other devices containing one or more dangerous goods (or residues thereof) that are an integral element of the article, necessary for its functioning and that cannot be removed for the purpose of transport. An inner packaging is not an article.

3.12.2 Such articles may in addition contain batteries. Lithium batteries that are integral to the article shall be of a type proven to meet the testing requirements of the UN Manual of Tests and Criteria, part III, sub-section 38.3, except when otherwise specified by these Regulations (e.g. for articles containing pre-production prototype lithium batteries or for a small production run, consisting of not more than 100 such batteries).

3.12.3 This section does not apply to articles for which a more specific proper shipping name already exists in the List of Dangerous Goods, Table 4.2.

3.12.4 This section does not apply to dangerous goods of Class 1, Division 6.2, Class 7 or radioactive material contained in articles. However, this section applies to articles containing explosives which are excluded from Class 1 in accordance with 3.1.7.4.

3.12.5 Articles containing dangerous goods must be assigned to the appropriate class or division determined by the hazards present using, where applicable, the Precedence of Hazards table in 3.10 for each of the dangerous goods contained in the article. If dangerous goods classified as Class 9 are contained within the

article, all other dangerous goods present in the article must be considered to present a higher hazard.

3.12.6 Subsidiary hazards must be representative of the primary hazard posed by the other dangerous goods contained within the article or they must be the subsidiary hazard(s) identified in Column C in Table 4.2 when only one dangerous goods is present in the article. If the article contains more than one dangerous goods and these could react dangerously with one another during transport, each of the dangerous goods must be enclosed separately (see 5.0.2.11(a)).









